Kontaktunabhängige elektrische Charakterisierung organischer Dünnschicht-Feldeffecktransistoren mittels der Gated Van-der-Pauw-Methode

Contact-Independent Electrical Characterization of Organic Thin-Film Field-Effect Transistors Using the Gated Van-der-Pauw Method

Bachelor Thesis

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ABSTRACT

Organic semiconductors are promising materials for lightweight, flexible, and large-area electronic devices. Their performance, however, cannot be described by material properties alone. In organic thin-film transistors, measured currents are shaped by the semiconductor, the device geometry, the contacts, the dielectric interface, and time-dependent charging processes. This makes mobility extraction intrinsically model-dependent and motivates measurement strategies that separate sheet transport in the semiconductor from parasitic device effects.

This thesis investigates this problem using bottom-gate C8-BTBT organic field-effect transistors and a current-driven gated van der Pauw approach. The work connects fabrication, device-quality checks, conventional transistor measurements, four-terminal sheet-conductance analysis, and transient circuit tests. The central aim is not only to demonstrate field-effect operation, but to evaluate how reliably transport parameters can be obtained when contacts, leakage paths, traps, and bias history may influence the electrical response.

Conventional current–voltage measurements confirm that the fabricated devices operate as *p*-type organic transistors with clear gate modulation and low gate leakage. At the same time, their gradual turn-on, finite background resistance, and nonideal transfer characteristics show that two-terminal measurements remain strongly affected by contact and injection effects. The gated van der Pauw measurements reduce this ambiguity by probing the accumulated film through a four-terminal geometry. The resulting sheet conductance is reproducible for different imposed currents and yields an effective thin-film mobility of about $3.1\,\mathrm{cm^2/(V\,s)}$.

Transient and amplifier measurements further show that the device response is not purely static. Slow relaxation and circuit time constants affect the dynamic behaviour and must be considered when interpreting transfer curves or designing organic transistor circuits. Overall, the thesis demonstrates that reliable characterization of organic field-effect devices requires a combination of careful fabrication control, contact-independent transport analysis, and awareness of time-dependent effects. The gated van der Pauw method is therefore shown to be a valuable tool for extracting robust transport parameters from organic thin-film transistors.

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CHAPTER 1 INTRODUCTION

1.1 Introduction

Organic semiconductors provide a route to electronic devices that differ in important ways from conventional inorganic semiconductor technology. They are soft molecular materials, can often be processed under comparatively mild conditions, and are suitable for deposition over large areas and on lightweight or mechanically flexible substrates. These features make them attractive for future display technologies, including foldable mobile devices and large-area screens that can be bent or rolled. At the same time, their electronic properties can be influenced by molecular design, film growth, interfaces, and processing conditions. Organic semiconductors are therefore relevant not only for flexible displays, but also for sensors, low-cost integrated circuits, and optoelectronic devices [1, 2].

This flexibility in materials and processing is also one of the reasons why organic electronic devices are difficult to characterize quantitatively. Charge transport is affected by processes on several length scales: molecular packing and localized states determine the local transport landscape, while grain boundaries, interfaces, contacts, and device geometry shape the macroscopic current. Charge-carrier mobility is commonly used to compare materials and devices, but it is not a quantity that is measured directly. It is extracted from electrical data with the help of a transport model. The resulting value therefore depends on the assumptions of that model, on the voltage range used for the extraction, and on nonidealities of the device itself.

Organic field-effect transistors are widely used for such mobility measurements. In a conventional two-terminal measurement, however, the measured voltage includes both the voltage drop across the semiconductor channel and the voltage losses at the metal–semiconductor contacts. The gradual-channel model usually applied to transfer curves assumes that these contact contributions are small compared with the channel resistance. This assumption can be problematic for organic thin films. Charge injection at the metal–organic interface depends on energetic alignment, local morphology, trap density, and gate bias [3]. As a consequence, the slope of a measured transfer characteristic may reflect changes in both the channel and the contacts. If the ideal transistor model is then applied without further checks, the extracted mobility can describe an apparent device response rather than the transport in the semiconductor film. In severe cases, conventional current–voltage analysis has been shown to overestimate mobility by one order of magnitude or more [4].

The gated van der Pauw method is one way to reduce this ambiguity. It combines field-effect modulation with a four-terminal sheet-conductance measurement. A current is driven through one pair of contacts, while a separate pair of nominally non-injecting contacts senses the voltage. The voltage drop at the current-injecting contacts is therefore not included directly in the measured van der Pauw resistance. By sweeping the gate voltage, the carrier density in the accumulation layer is changed, and the corresponding sheet conductance can be used to determine a threshold voltage and an effective thin-film mobility that is less sensitive to contact voltage losses.

The word contact-independent must nevertheless be used with care. The current still has to enter and leave the organic semiconductor through metallic contacts, and poor injection can limit the accessible measurement range or require large driving voltages. The method only removes the contact voltage drops from the resistance used for the sheet-conductance extraction, provided that the imposed current is stable and the voltage probes draw negligible current. The extracted mobility also remains a macroscopic device parameter. It can include the influence of energetic disorder, traps, grain boundaries, and spatial inhomogeneities, and should not automatically be identified with an intrinsic molecular or intra-grain mobility.

The aim of this thesis is to characterize organic thin-film field-effect devices by combining conventional current–voltage measurements with the current-driven gated van der Pauw method. The work begins with the characterization of a prototype van der Pauw measurement setup and continues with the fabrication and analysis of molecular semiconductor devices prepared using physical vapor deposition, chemical vapor deposition, and spin coating. Special attention is paid to the local gate overdrive, to the dependence of the extracted quantities on the applied measurement current, and to contact and transient effects that can influence the measured response. The measurements are therefore treated not as isolated electrical curves, but as tests of the conditions under which reliable transport parameters can be obtained.

1.2 Status of Research

The classical van der Pauw method is a four-terminal technique for determining the sheet resistance of thin conducting films without using a Hall-bar geometry. Its strength lies in the separation of current injection and voltage sensing. Ideally, the voltage contacts draw no current, so the voltage losses at the current-injecting contacts do not enter directly into the measured sheet resistance. This makes the method especially useful for thin films in which the contact resistance is large, difficult to determine, or strongly dependent on the measurement conditions.

Such contact effects are a central issue in organic field-effect transistors. At metal–organic interfaces, charge injection is influenced by energetic alignment, interfacial states, local morphology, and gate bias. Conventional two-terminal transfer curves can therefore contain both channel and contact contributions [3]. Bittle et al. showed that gate-dependent contact resistance can lead to a strong overestimation of the field-effect mobility when standard transistor formulas are applied directly to two-terminal data [4]. A high apparent mobility obtained from a transfer curve is therefore not sufficient evidence for fast transport in the semiconductor film.

The gated van der Pauw method extends the same four-terminal idea to a gated accumulation layer. Rolin et al. introduced the approach for organic semiconductor thin films and showed that the gate dependence of the sheet conductance can be used to extract a long-range accumulation-layer mobility with reduced influence from the injecting contacts [5]. In this geometry, the gate controls the carrier density in the film, the imposed current defines the transport path, and the transverse van der Pauw voltage gives access to the sheet conductance of the probed region.

Later studies have broadened this idea beyond the original material system. Bonafè et al. adapted gated van der Pauw concepts to electrolyte-gated organic mixed ionic–electronic conductors [6]. Lee et al. applied related methods to thin-film transistors for which contact resistance becomes more important as device dimensions are reduced [7]. A subsequent voltage-driven implementation showed that channel and contact contributions can be evaluated separately within one device [8]. Together, these studies underline the same point: reliable mobility extraction requires a measurement strategy that can distinguish sheet transport from contact-related voltage losses.

For a practical organic thin-film device, this separation is only meaningful if the measurement conditions are well controlled. The prototype setup must provide stable current injection, reliable voltage sensing, and sufficiently low leakage. The applied current has to be large enough to produce a measurable transverse voltage, but small enough to keep the conducting sheet close to a linear-response regime. Gate leakage must remain small compared with the channel current, and the extracted mobility should be reproducible for different current levels. If the conductance curves depend strongly on the selected current, the contact configuration, or the sweep history, the assumptions of a simple gated van der Pauw extraction are no longer fully satisfied. For this reason, the present thesis combines setup characterization, device fabrication, conventional current–voltage measurements, current-driven gated van der Pauw measurements, and additional transient/circuit tests.

1.3 Scientific Questions

The central question of this thesis is whether organic thin-film transistors can be fabricated and characterized in a way that separates the transport through the accumulated semiconductor sheet from contact-related artefacts. Demonstrating field-effect behaviour alone is not enough for this purpose. The more important question is how reliably mobility and threshold voltage can be extracted when contact resistance, charge injection, leakage, traps, and transient relaxation may all contribute to the measured current.

The first part of this question concerns the experimental platform. Before the gated van der Pauw method can be applied to molecular semiconductor devices, the prototype setup has to be checked for stable current driving and reliable voltage detection. On the fabrication side, the devices have to be produced in a layer-by-layer process that combines physical vapor deposition of the molecular semiconductor and metal contacts, chemical vapor deposition of the parylene-N insulator, and spin coating of the planarization layer. The completed structures must then show low gate leakage, reproducible field-effect modulation, and ordered transfer and output characteristics. Without these basic conditions, neither conventional transistor analysis nor gated van der Pauw analysis would give meaningful transport parameters.

The second part concerns the role of the contacts. In a two-terminal measurement, the same electrodes inject current and define the measured voltage. Injection barriers and contact resistance can therefore change the slope of a transfer curve and distort the mobility extracted from standard FET relations. The gated van der Pauw measurement tests whether a sheet conductance obtained from non-injecting voltage probes gives a more robust result. If measurements at different imposed currents agree, this is an important indication that the extracted mobility reflects the accumulated film rather than a contact-limited current path.

The third part concerns dynamic effects. Organic semiconductors often contain localized states that can trap and release carriers over long time scales. Such processes can influence threshold-voltage extraction, hysteresis, and circuit response even when a static transfer curve appears smooth [9, 10]. DLTS-type transient measurements and a simple amplifier test are therefore used to examine whether slow relaxation and external circuit elements affect the device response.

The work is guided by three working hypotheses. First, in a suitable accumulation regime, the sheet conductance measured by the gated van der Pauw method should increase approximately linearly with effective gate bias, and the mobility extracted from this slope should be similar for different sufficiently small measurement currents. Second, conventional two-terminal analysis should be more sensitive to contact effects than the four-terminal gated van der Pauw analysis. Evidence for this would include a finite background resistance, nonideal transfer or output curves, or a discrepancy between contact-sensitive and contact-independent interpretations. Third, transient measurements should show whether slow trap-related relaxation is relevant for the fabricated devices and whether it has to be considered when interpreting nominally static measurements.

The following chapters test these hypotheses through controlled comparisons. The key checks are the stability of the measurement setup, the integrity of the fabricated device stack, the magnitude of the gate leakage, the shape of the transfer and output curves, the linearity of the gated van der Pauw conductance, the current independence of the extracted mobility, and the presence or absence of slow transient relaxation. In this way, the thesis connects setup characterization, fabrication, measurement geometry, and data analysis to the question of how reliably transport parameters can be obtained from organic thin-film transistors.

CHAPTER 2

THEORETICAL AND TECHNICAL BACKGROUNDS

2.1 Semiconductor Physics in Equilibrium

This section introduces the equilibrium physics of semiconductors. It first outlines the formation of allowed and forbidden energy bands and uses this picture to classify materials as conductors, semiconductors, or insulators. It then summarizes charge-carrier statistics and basic transport mechanisms before discussing Schottky and Ohmic metal-semiconductor contacts. Finally, the section introduces the metal-insulator-semiconductor (MIS) capacitor as a key structure for understanding field-effect devices.

2.1.1 Energy Bands and Classification of Materials

In an isolated atom, electrons occupy discrete energy levels. In a solid, however, a large number of atoms are arranged close to each other. Due to the interaction between neighboring atoms and the Pauli exclusion principle, the discrete atomic energy levels split into a large number of closely spaced states. These states form quasi-continuous energy bands, while energy regions with no available electronic states are called band gaps [11, 12].

For the description of electrical conduction, the most important bands are the valence band and the conduction band. The valence band is the highest band that is completely or almost completely occupied at low temperature, whereas the conduction band contains mobile electrons that can contribute to electrical transport. The energy difference between the conduction-band minimum E_C and the valence-band maximum E_V is called the band gap,

$$E_\mathrm{g} = E_\mathrm{C} - E_\mathrm{V}. \tag{2.1}$$

The occupation probability of an electronic state with energy E in thermal equilibrium is described by the Fermi–Dirac distribution

$$f(E) = \frac{1}{1 + \exp\left(\frac{E - E_\mathrm{F}}{k_\mathrm{B}T}\right)}, \tag{2.2}$$

where E_F is the Fermi energy, k_B is the Boltzmann constant, and T is the absolute temperature. The Fermi energy represents the electrochemical potential of the electron system in thermal equilibrium. At $E = E_\mathrm{F}$, the occupation probability is equal to 1/2 [12].

Depending on their electrical resistivity and band structure, solids can be classified into metals, semiconductors, and insulators. Semiconductors typically have a finite band gap below about 4 eV, whereas insulators usually have larger band gaps. The boundaries between these classes are not perfectly sharp, since the resistivity depends on temperature, purity, crystal structure and doping concentration. Nevertheless, the orders of magnitude listed in Table 2.1 are useful for a qualitative classification [11, 12].

Material class	**Typical resistivity** ρ	**Band gap** E_g	**Examples**
Metal	10^{-8}–$10^{-6}\,\Omega\,\mathrm{m}$	Small or overlap	Cu, Ag, Au, Al
Semiconductor	10^{-4}–$10^{7}\,\Omega\,\mathrm{m}$	$0 < E_\mathrm{g} < 4\,\mathrm{eV}$	Si, Ge, GaAs, pentacene, C8-BTBT
Insulator	$> 10^{7}\,\Omega\,\mathrm{m}$, often up to $10^{16}\,\Omega\,\mathrm{m}$ or higher	$E_\mathrm{g} > 4\,\mathrm{eV}$	SiO_2, glass, Al_2O_3, polystyrene

Table 2.1: Classification of materials according to typical electrical resistivity and approximate band-gap range.

In metals, electrons can easily occupy nearby empty states because the Fermi level lies inside an allowed energy band. Therefore, metals show high electrical conductivity and low resistivity [11, 12].

In semiconductors, the band gap is finite but sufficiently small that charge carriers can be generated thermally or by external excitation. Electrons in the conduction band and holes in the valence band can both contribute to charge transport. In addition, the carrier density can be controlled by doping or by an external electric field, which is the physical basis of field-effect devices [12].

In insulators, the band gap is large and the number of thermally generated charge carriers at room temperature is negligible. Therefore, their electrical resistivity is very high [11, 12].

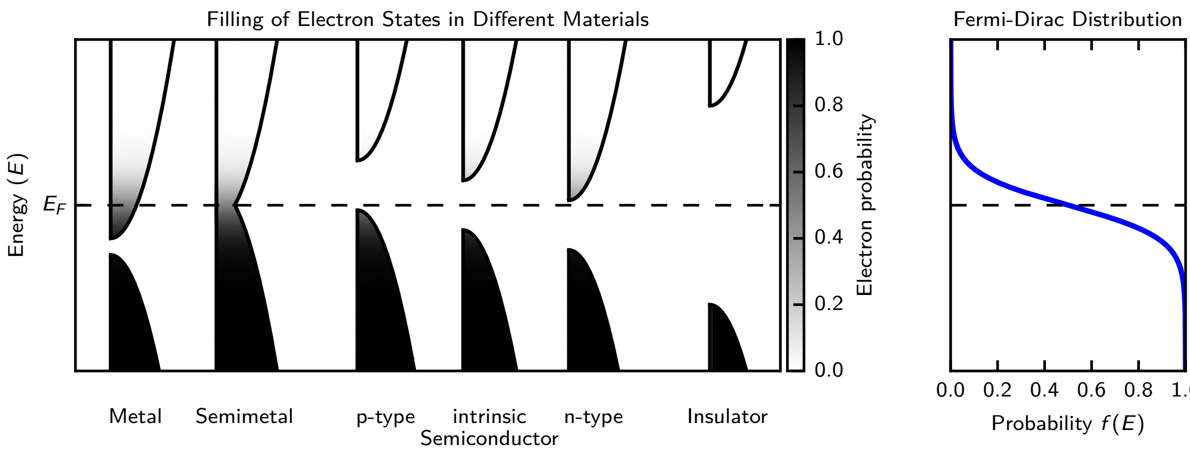

Figure 2.1: Electron occupancy of allowed energy bands in metals, semimetals, semiconductors, and insulators at temperature $T > 0\,\mathrm{K}$. Shading indicates Fermi–Dirac occupation from filled states (black) to empty states (white).

The same terminology is often used for organic semiconductors, although the microscopic origin of the relevant electronic states is different. Instead of ideal crystal bands, organic semiconductors are commonly described in terms of molecular orbitals. The highest occupied molecular orbital (HOMO) is analogous to the valence band, while the lowest unoccupied molecular orbital (LUMO) is analogous to the conduction band. In p-type organic semiconductors such as C8-BTBT, charge transport is mainly associated with holes in HOMO-derived transport states [1].

2.1.2 Charge Carriers Statistics and Transport

The electrical conductivity of a semiconductor is determined by the number of mobile charge carriers and by their ability to move through the material. The mobile carriers are electrons in the conduction band and holes in the valence band. Their concentrations are obtained from the density of available electronic states and from the occupation probability of these states [12].

The electron density in the conduction band is obtained by multiplying the available density of states by the occupation probability and integrating over all conduction-band energies. For a non-degenerate semiconductor, this expression can be written in the Boltzmann approximation as [12]

$$n = \int_{E_\mathrm{C}}^{\infty} D_\mathrm{C}(E) f(E)\,\mathrm{d}E \approx N_\mathrm{C} \exp\left(-\frac{E_\mathrm{C} - E_\mathrm{F}}{k_\mathrm{B}T}\right), \tag{2.3}$$

where $D_\mathrm{C}(E)$ is the density of states in the conduction band and N_C is the effective density of states in the conduction band. Correspondingly, the hole density in the valence band is

$$p = \int_{-\infty}^{E_\mathrm{V}} D_\mathrm{V}(E)\,[1 - f(E)]\,\mathrm{d}E \approx N_\mathrm{V} \exp\left(-\frac{E_\mathrm{F} - E_\mathrm{V}}{k_\mathrm{B}T}\right), \tag{2.4}$$

where $D_\mathrm{V}(E)$ is the density of states in the valence band and N_V is the effective density of states in the valence band. The factor $1 - f(E)$ describes the probability that an electronic state in the valence band is empty. These equations show that the electron density increases when the Fermi level moves closer to the conduction band, whereas the hole density increases when the Fermi level moves closer to the valence band [12].

For an intrinsic semiconductor, the electron and hole concentrations are equal,

$$n = p = n_i = \sqrt{N_\mathrm{C} N_\mathrm{V}} \exp\left(-\frac{E_\mathrm{g}}{2k_\mathrm{B}T}\right), \tag{2.5}$$

where n_i is the intrinsic carrier concentration [12].

In thermal equilibrium, the product of electron and hole density is given by the mass action law [12]

$$np = n_i^2. \tag{2.6}$$

The carrier concentrations can also be expressed relative to the intrinsic Fermi level E_Fi [12],

$$n = n_i \exp\left(\frac{E_\mathrm{F} - E_\mathrm{Fi}}{k_\mathrm{B}T}\right), \tag{2.7}$$

$$p = n_i \exp\left(\frac{E_\mathrm{Fi} - E_\mathrm{F}}{k_\mathrm{B}T}\right). \tag{2.8}$$

For parabolic bands and similar effective masses of electron and hole, the intrinsic Fermi level is approximately

$$E_\mathrm{Fi} \approx \frac{1}{2}\left(E_\mathrm{V} + E_\mathrm{C}\right), \tag{2.9}$$

which lies close to the middle of the band gap [12].

This relation is valid for a non-degenerate semiconductor in thermal equilibrium. If the semiconductor is doped, the Fermi level shifts away from the intrinsic level, as illustrated in Figure 2.2. In a p-type semiconductor, acceptor atoms increase the hole concentration and shift the Fermi level closer to the valence band. For a strongly doped p-type semiconductor with complete ionization of acceptors, the carrier densities are approximately [12]

$$p \approx N_\mathrm{A}, \qquad n = \frac{n_i^2}{p} \ll p. \tag{2.10}$$

In an n-type semiconductor, donor atoms increase the electron concentration and shift the Fermi level closer to the conduction band. Similarly, for a strongly doped n-type semiconductor with complete ionization of donors, the carrier densities are approximately [12]

$$n \approx N_\mathrm{D}, \qquad p = \frac{n_i^2}{n} \ll n. \tag{2.11}$$

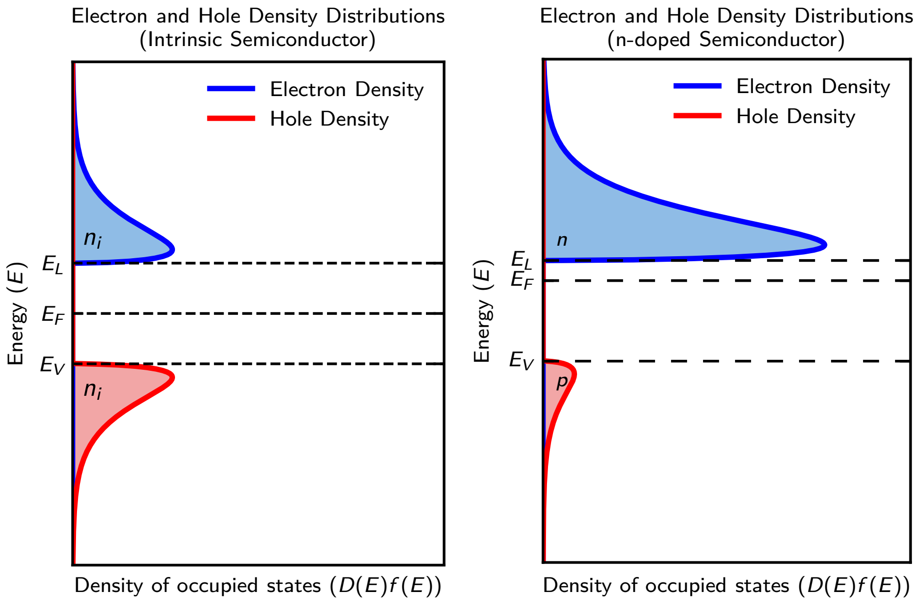

Figure 2.2: Carrier concentrations in intrinsic (left) and p-type (right) semiconductors. In an intrinsic semiconductor, electron and hole concentrations are equal, $n = p = n_i$. Doping shifts the Fermi level and changes the majority and minority carrier densities.

Charge transport in semiconductors is caused by drift and diffusion. In the Drude model, mobile charge carriers are accelerated by an electric field and lose momentum through scattering events characterized by an average relaxation time τ. This leads to a drift velocity proportional to the electric field, with the proportionality described by the mobility, $\mu = q\tau/m^*$, where m^* is the effective mass [12]. Drift transport therefore occurs when an electric field acts on mobile charge carriers. The electric field is related to the electrostatic potential by

$$\mathbf{E} = -\nabla\varphi. \tag{2.12}$$

The spatial distribution of charge carriers is coupled to the electrostatic potential through Poisson's equation [12],

$$\nabla \cdot (-\epsilon_0\epsilon_r\nabla\varphi) = \rho = e\left(p - n + N_\mathrm{D}^+ - N_\mathrm{A}^-\right), \tag{2.13}$$

where ϵ_0 is the vacuum permittivity, ϵ_r is the relative permittivity, e is the elementary charge, and N_{D}^{+} and N_{A}^{-} are the ionized donor and acceptor concentrations, respectively. The quantities μ_n and μ_p denote the electron and hole mobilities, which describe how fast charge carriers move in response to an applied electric field. Diffusion transport occurs when the carrier concentration is spatially inhomogeneous, causing charge carriers to move from regions of high concentration to regions of low concentration. Combining drift and diffusion, the total current density can be written as [12]

$$\mathbf{J} = q\left(n\mu_n + p\mu_p\right)\mathbf{E} + q\left(D_n\nabla n - D_p\nabla p\right). \tag{2.14}$$

In steady state, charge conservation requires that the current density is divergence-free,

$$\nabla \cdot \mathbf{J} = 0. \tag{2.15}$$

For a homogeneous semiconductor, where concentration gradients can be neglected, the current density is dominated by drift. The electrical conductivity is then

$$\sigma = q\left(n\mu_n + p\mu_p\right). \tag{2.16}$$

Compared with an intrinsic semiconductor, a doped semiconductor usually has a much larger conductivity because doping increases the majority carrier concentration by many orders of magnitude. Since the conductivity is proportional to the carrier density, this increase can dominate even if the mobility is reduced by impurity scattering [12]. For a p-type organic semiconductor such as C8-BTBT [1], in which holes are the majority carriers and the electron contribution is negligible, this simplifies to

$$\sigma \approx qp\mu_p. \tag{2.17}$$

In thermal equilibrium, no net current flows. This does not mean that electrons and holes are immobile. Rather, microscopic motion is still present, but drift and diffusion currents balance each other exactly. This balance is expressed by the Einstein relation [12],

$$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = \frac{k_{\mathrm{B}}T}{q}. \tag{2.18}$$

In organic semiconductors, the same macroscopic quantities, such as carrier density, mobility and conductivity, are used to describe charge transport. However, the microscopic transport mechanisms can differ from those in ideal inorganic crystals. Due to weak van der Waals coupling, energetic disorder, grain boundaries and trap states, charge transport in organic thin films may involve both band-like motion and hopping between localized states. In polycrystalline organic semiconductors, grain boundaries and traps can reduce the effective mobility measured over macroscopic distances. Therefore, the mobility extracted from electrical measurements should be understood as an effective transport parameter of the investigated thin film [1].

2.1.3 Metal-Semiconductor Contacts

Metal–semiconductor contacts determine how efficiently carriers enter and leave the semiconductor. Even if transport in the channel is described by drift and diffusion, the measured current can be limited by injection barriers at the electrodes. The two limiting cases are rectifying Schottky contacts and nearly barrier-free Ohmic contacts [12, 13].

Before contact, the metal and semiconductor have independent Fermi levels. After contact, charge transfer aligns the Fermi levels and bends the semiconductor bands near the interface, as shown in Figure 2.3. For a p-type semiconductor, a low metal work function relative to the semiconductor work function depletes holes at the interface and produces a Schottky barrier. In the ideal Schottky–Mott picture this rectifying case is expected when

$$\phi_{\mathrm{m}} < \phi_{\mathrm{s}}, \tag{2.19}$$

and the approximate hole barrier height is

$$e\phi_{\mathrm{B}} = E_{\mathrm{g}} - e(\phi_{\mathrm{m}} - \chi). \tag{2.20}$$

Thus, higher-work-function electrodes reduce the barrier for hole injection, whereas lower-work-function electrodes increase the contact resistance [12].

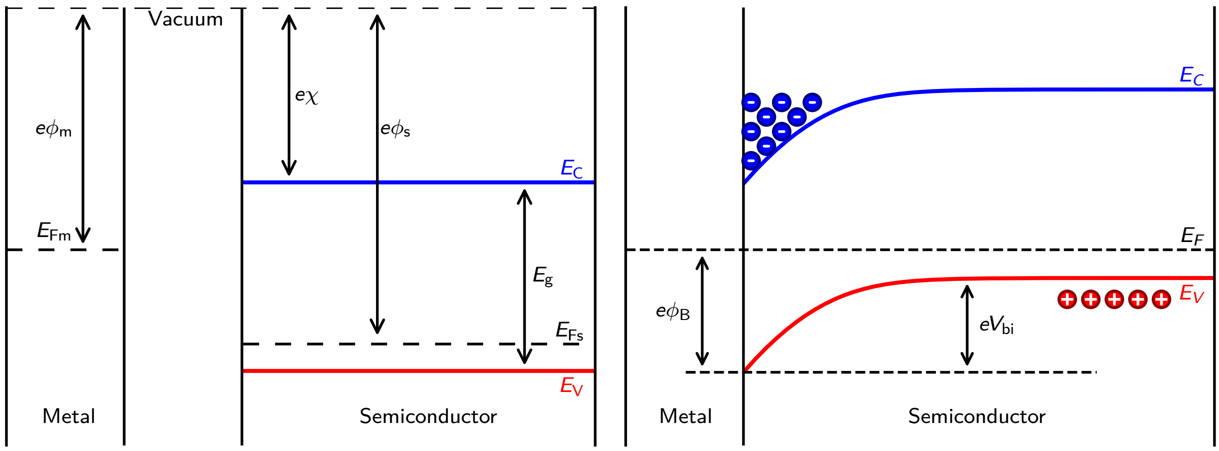

Figure 2.3: Schottky barrier formation at a metal–p-type semiconductor contact. Band alignment of a metal and a p-type semiconductor before (left) and after contact (right). Fermi-level alignment causes band bending in the semiconductor and can form a Schottky barrier for hole transport.

For C8-BTBT devices, the HOMO or ionization energy is about 5.4 eV, while chromium has a lower work function of roughly 4.5 eV. A direct Cr/C8-BTBT interface is therefore expected to form a hole-injection barrier. High-work-function MoO_x or MoO_3 interlayers can reduce this barrier and are commonly used to improve hole injection in organic field-effect transistors [14, 15, 16, 17].

The band diagram also explains the bias dependence of contact behavior. In a Schottky contact, band bending creates a depletion region and a bias-dependent barrier: forward bias lowers the barrier, while reverse bias widens the depletion region and suppresses injection. In contrast, an Ohmic contact has a small or tunnelable injection barrier, so the current–voltage relation is approximately linear at small bias. For p-type semiconductors this is favored when $\phi_{\mathrm{M}} \gtrsim \phi_{\mathrm{S}}$, although interface dipoles, chemical reactions, disorder, and traps often modify the ideal work-function picture [12, 13].

Several microscopic processes can contribute to carrier transfer across the interface. If the barrier is relatively high and wide, carriers must be thermally excited over it; this thermionic-emission process is strongly temperature dependent and gives rectifying behavior. Once carriers enter the semiconductor, they are transported away from the interface by drift in the local electric field and by diffusion caused by carrier-concentration gradients. If the barrier is thin, for example due to strong band bending, heavy doping, or an interfacial oxide, carriers can also tunnel through it. Real organic contacts often combine these mechanisms with hopping through localized interfacial states, so the observed contact resistance reflects both the injection barrier and the near-interface transport region [12, 1].

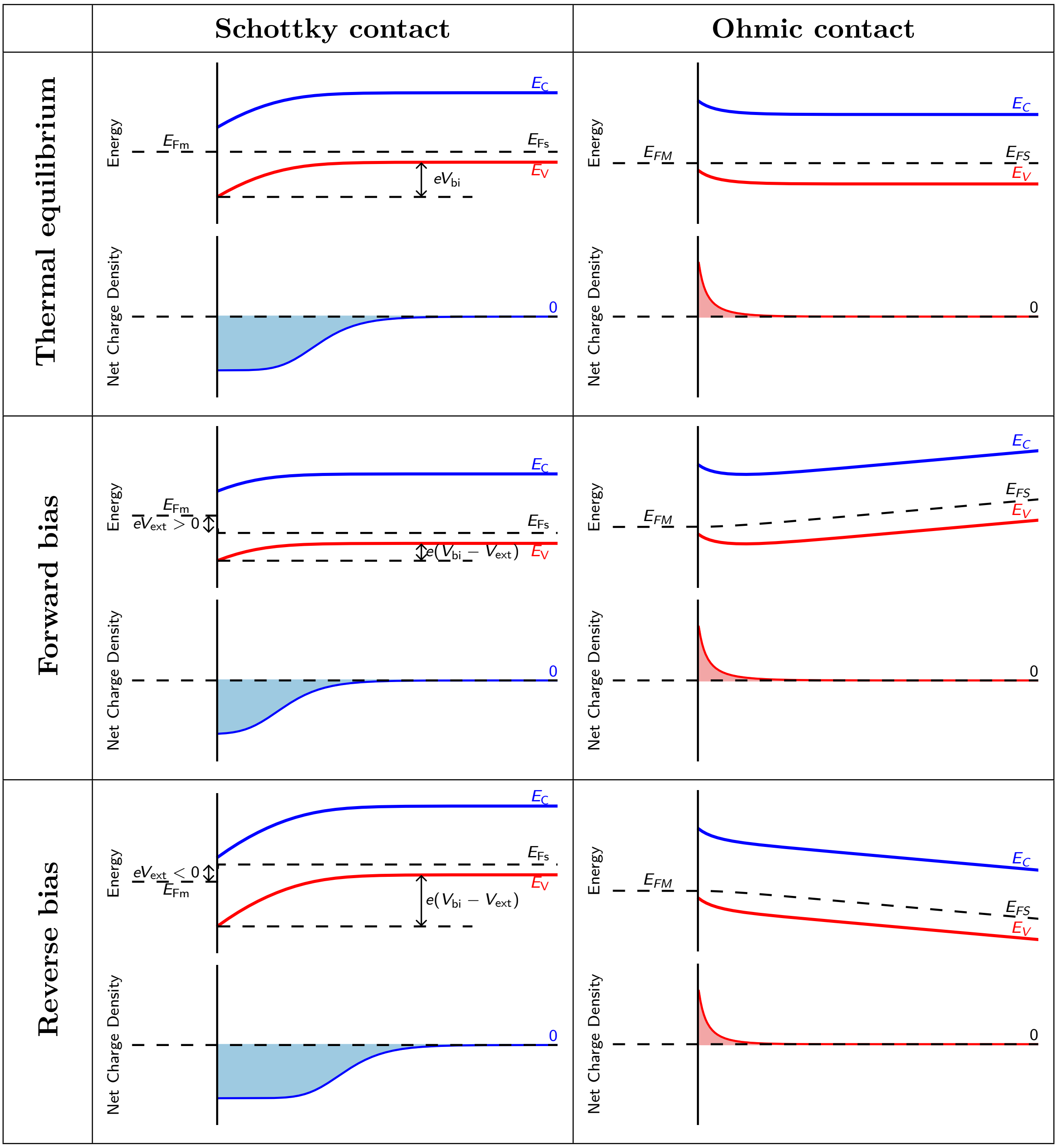

Figure 2.4: Band diagrams and corresponding net charge-density profiles for metal–semiconductor contacts under equilibrium, forward-bias, and reverse-bias conditions.

2.1.4 Metal-Insulator-Semiconductor Capacitor

A metal–insulator–semiconductor (MIS) capacitor consists of a metal gate, an insulating dielectric, and a semiconductor. Because the dielectric blocks direct DC current, the gate voltage controls the semiconductor mainly through the electric field across the insulator. This field changes the surface potential and therefore the carrier density at the semiconductor–insulator interface. The MIS capacitor is thus the basic electrostatic model for field-effect devices [12, 18].

In a band diagram, flat-band conditions correspond to nearly horizontal semiconductor bands and no net space-charge region at the surface. Applying a gate voltage bends the bands near the interface. For a p-type semiconductor, negative gate bias attracts holes and produces accumulation, while positive gate bias repels holes and produces depletion. Strong positive bias can lead to inversion in ideal inorganic semiconductors, but in many p-type organic semiconductors the practically relevant regimes are mainly accumulation and depletion [19, 1].

The gate dielectric is described by its geometric area capacitance,

$$C_\mathrm{i} = \frac{\epsilon_0 \epsilon_\mathrm{i}}{d_\mathrm{i}}, \tag{2.21}$$

so the induced sheet charge density in accumulation is approximately

$$p_\mathrm{2D} \approx \frac{C_\mathrm{i} |V_\mathrm{G} - V_\mathrm{th}|}{e}. \tag{2.22}$$

Here V_th represents flat-band shifts, trap filling, and contact effects. The MIS picture therefore links gate voltage, band bending, interfacial carrier density, and field-effect transport in organic semiconductor devices [18, 19].

2.2 Organic Semiconductors

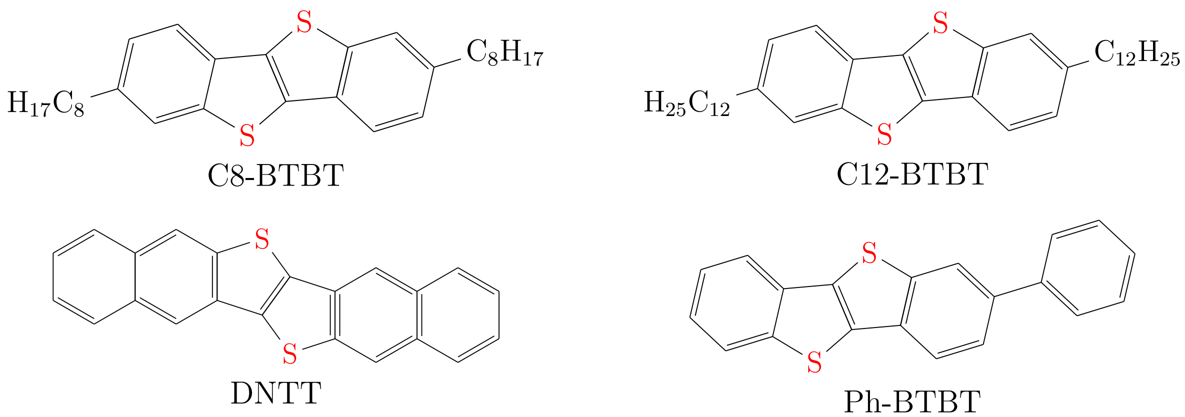

Figure 2.5: Chemical structures of common high-mobility small-molecule organic semiconductors: C8-BTBT, C12-BTBT, DNTT, and Ph-BTBT.

Organic semiconductors are carbon-based molecular or polymeric materials in which electronic transport is governed by conjugated π-electron systems. Unlike inorganic semiconductors, where atoms are connected by strong covalent bonds throughout a crystal lattice, organic semiconductors are usually held together by weaker intermolecular interactions. The resulting electronic coupling between molecules is comparatively small, so charge transport is highly sensitive to molecular packing, energetic disorder, defects, interfaces, and trap states [1, 20]. Representative high-mobility small-molecule organic semiconductors, including BTBT derivatives and DNTT, are shown in Figure 2.5 [15, 2].

For this thesis, the relevant physical picture is that the gate voltage and the contacts determine how many carriers are injected or accumulated, while the molecular solid determines how efficiently those carriers move. The following subsections summarize the molecular electronic structure, disorder and trap states, carrier distribution in field-effect geometries, and temperature-dependent transport mechanisms.

2.2.1 Molecular Structure and Electronic States

The frontier molecular orbitals dominate the electronic properties of organic semiconductors. The *highest occupied molecular orbital* (HOMO) is analogous to the valence band maximum of an inorganic semiconductor, while the *lowest unoccupied molecular orbital* (LUMO) is analogous to the conduction band minimum. Their energetic separation is commonly called the transport gap,

$$E_\mathrm{g} = E_\mathrm{LUMO} - E_\mathrm{HOMO}. \tag{2.23}$$

For p-type materials, hole transport is associated mainly with HOMO-derived states; for n-type materials, electron transport occurs through LUMO-derived states [1, 20].

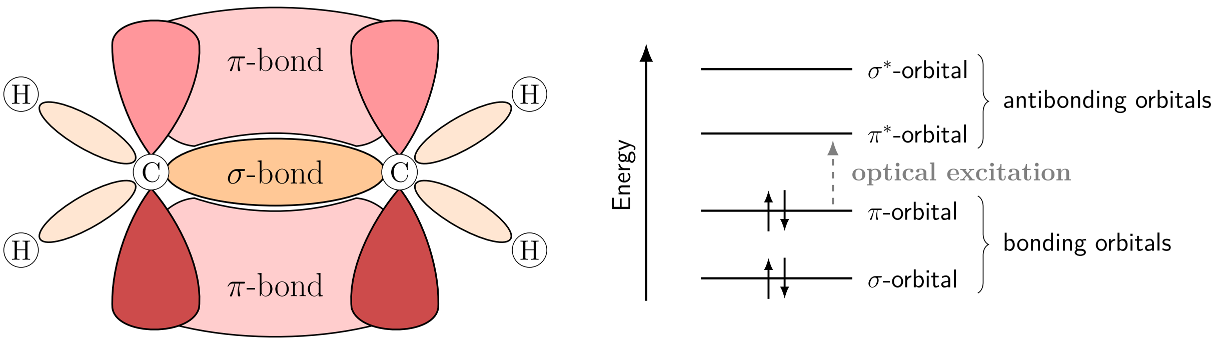

Figure 2.6: Sigma and pi bonding in ethene as the simplest conjugated pi-electron system. The left panel illustrates the spatial overlap of the σ and π bonds, while the right panel shows the corresponding bonding and antibonding orbital levels. The lowest optical excitation occurs from the bonding π orbital to the antibonding π^* orbital. Adapted from Ref. [21] and [12].

The origin of semiconducting behaviour is π-conjugation. In conjugated molecules, neighbouring carbon atoms are typically sp^2 hybridized, leaving p_z orbitals that overlap to form delocalized π orbitals. Within one molecule this delocalization can be strong, but transport through a solid also requires overlap between orbitals on neighbouring molecules. Because this intermolecular overlap is weak, small changes in molecular packing, tilt angle, thermal motion, or grain orientation can strongly change the electronic transfer integral and therefore the mobility [20].

The basic bonding picture is illustrated in Figure 2.6. Ethene is the simplest example of a conjugated π-electron system: the σ bond forms the molecular backbone, while the p_z orbitals combine into bonding π and antibonding π^* orbitals. In longer conjugated molecules, many such orbitals split into a set of occupied and unoccupied levels. The lowest optical excitation is then typically associated with a transition from the bonding π-derived HOMO to the antibonding π^*-derived LUMO [1].

Organic semiconductors are often classified as small molecules or conjugated polymers. Small molecules can form crystals or polycrystalline films with grains and grain boundaries, whereas polymers usually contain both ordered and disordered chain segments. In both classes, improved molecular order generally increases orbital overlap and reduces energetic disorder, but real thin films remain strongly influenced by processing conditions and interfaces [1, 2].

2.2.2 Energetic Disorder and Trap States

A real organic semiconductor thin film is not an ideal periodic crystal. Variations in molecular position, orientation, grain structure, impurities, interface roughness, and dielectric dipoles create local shifts of the HOMO and LUMO energies. Therefore, the transport states are not represented by sharp band edges. A common approximation for disordered organic semiconductors is a Gaussian density of states (DOS),

$$g(E) = \frac{N_0}{\sqrt{2\pi}\sigma_{\mathrm{DOS}}} \exp\left[-\frac{(E-E_0)^2}{2\sigma_{\mathrm{DOS}}^2}\right], \tag{2.24}$$

where N_0 is the total density of localized states, E_0 is the centre of the distribution, and σ_{DOS} describes the energetic disorder. Tail states and additional deep traps can be superimposed on this Gaussian background. This Gaussian-disorder picture is widely used to describe hopping transport in organic semiconductors [22, 1, 23].

Trap states are localized states that temporarily immobilize charge carriers. For a p-type semiconductor, hole traps reduce the density of mobile holes; for an n-type semiconductor, electron traps play the corresponding role. The total carrier density can therefore be separated into mobile and trapped parts,

$$n_{\mathrm{tot}} = n_{\mathrm{free}} + n_{\mathrm{trap}}. \tag{2.25}$$

Only the mobile part contributes directly to the conductivity,

$$\sigma = q n_{\mathrm{free}} \mu, \tag{2.26}$$

where q is the elementary charge and μ is the effective mobility [1, 9].

Shallow traps lie close to the transport level and can release carriers by thermal excitation. Deep traps are farther from the transport level and can store charge for longer times. Deep or slowly exchanging traps can cause threshold voltage shifts, hysteresis, bias-stress effects, and memory during cooling and heating cycles. Consequently, temperature-dependent measurements probe not only a mobility change, but also a redistribution of carriers between mobile and trapped states [9, 10].

2.2.3 Charge Carrier Distribution

Because many organic semiconductors have relatively large transport gaps, their intrinsic thermally generated carrier density is small. In field-effect devices, the relevant carrier density is mainly set by contact injection and by the gate electric field. For a p-type semiconductor, a negative gate voltage attracts holes to the semiconductor–dielectric interface and forms an accumulation layer; for an n-type semiconductor, a positive gate voltage accumulates electrons [2, 1].

In the gradual-channel approximation, the induced sheet charge density can be written as

$$Q = C_{\mathrm{i}} \left(V_{\mathrm{G}} - V_{\mathrm{T}} - V \right), \tag{2.27}$$

where C_{i} is the gate-insulator capacitance per unit area, V_{G} is the gate voltage, V_{T} is an effective threshold voltage, and V is the local channel potential. The corresponding two-dimensional carrier density is

$$n_{2\mathrm{D}} = \frac{|Q|}{q} = \frac{C_{\mathrm{i}}}{q} \left| V_{\mathrm{G}} - V_{\mathrm{T}} - V \right|. \tag{2.28}$$

These relations show that the carrier density is not fixed only by the externally applied gate voltage; it also varies along the channel through the local potential. This point is important for transistor measurements and for gated van der Pauw geometries, where the measured conductance corresponds to a specific probed region of the film [2, 5].

Microscopically, the carrier density follows from the Gaussian DOS and the Fermi–Dirac occupation. For electron transport through LUMO-derived states,

$$n = \int_{-\infty}^{\infty} g_{\mathrm{LUMO}}(E) f(E, E_{\mathrm{F}}, T)\, \mathrm{d}E, \tag{2.29}$$

whereas the hole density in HOMO-derived states is

$$p = \int_{-\infty}^{\infty} g_{\mathrm{HOMO}}(E) \left[1 - f(E, E_{\mathrm{F}}, T) \right] \mathrm{d}E. \tag{2.30}$$

Here g_{HOMO} and g_{LUMO} are often modeled by Gaussian distributions of the form of Equation 2.24. Shifting the Fermi level by the gate field therefore changes both the number of mobile carriers and the occupation of localized tail and trap states [24, 23].

2.2.4 Charge Carrier Transport Mechanisms

Charge transport in organic semiconductors is not described by a single universal mechanism. The dominant process depends on molecular order, temperature, energetic disorder, carrier density, and contact quality. In highly ordered single crystals or well-ordered films, carriers can show band-like transport, in which the mobility decreases with increasing temperature because molecular vibrations disturb intermolecular electronic coupling. A common empirical form is

$$\mu(T) \propto T^{-n}, \tag{2.31}$$

where $n > 0$ [25, 20].

In more disordered thin films, transport is often described as hopping between localized states. A hop depends on the distance between sites, their energy difference, the electric field, and the thermal energy available to overcome local barriers. Thermally activated hopping often gives a mobility that increases with temperature and can be approximated by an Arrhenius relation,

$$\mu(T) = \mu_0 \exp\left(-\frac{E_{\mathrm{a}}}{k_{\mathrm{B}} T} \right), \tag{2.32}$$

where μ_0 is a prefactor and E_{a} is an effective activation energy. This activation energy can include contributions from energetic disorder, trap release, grain-boundary barriers, and contact injection [22, 1].

In a band-transport picture, the mobility is limited by scattering processes that reduce the average momentum-relaxation time τ. The Drude relation $\mu = q\tau/m^*$ therefore shows that stronger scattering directly lowers the mobility. Two important limiting mechanisms are phonon scattering and ionized impurity scattering. Phonon scattering is caused by lattice vibrations or, in organic crystals, thermally excited molecular motions that modulate the intermolecular transfer integrals. As the temperature increases, these vibrations become stronger and the phonon-limited mobility typically decreases. In many crystalline semiconductors this trend is written approximately as $\mu_{\mathrm{ph}} \propto T^{-3/2}$, although the exact exponent depends on the material and dominant phonon mode. Ionized impurity scattering originates from the Coulomb potential of charged dopants, trapped charges, or fixed interface charges. It becomes stronger when the concentration of charged scattering centers increases and is often approximated by $\mu_{\mathrm{ii}} \propto T^{3/2}/N_{\mathrm{i}}$, where N_{i} is the ionized impurity concentration. If several independent scattering processes are present, their contributions can be combined using Matthiessen's rule,

$$\frac{1}{\mu_{\mathrm{tot}}} = \frac{1}{\mu_{\mathrm{ph}}} + \frac{1}{\mu_{\mathrm{ii}}} + \cdots . \tag{2.33}$$

This relation shows that the smallest partial mobility, corresponding to the strongest scattering channel, dominates the measured mobility. In organic thin films these simple power laws should be interpreted qualitatively, because energetic disorder, trapping, grain boundaries, and hopping transport often contribute at the same time [12, 1, 20].

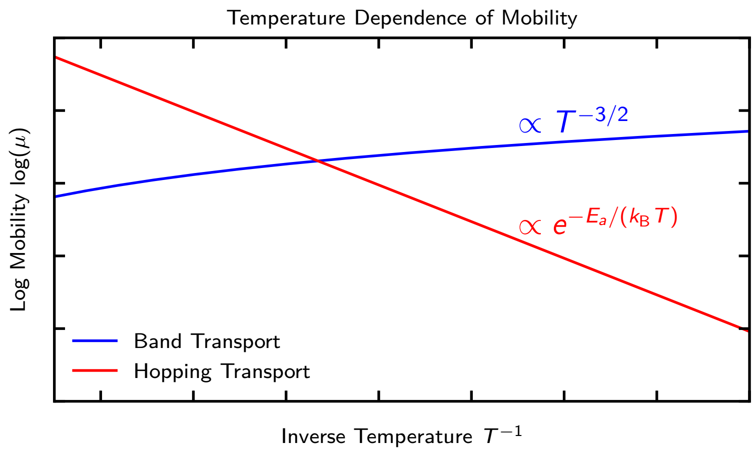

Figure 2.7: Schematic temperature dependence of charge transport in organic semiconductors. Thermally activated hopping typically becomes more efficient at higher temperature, whereas band-like transport can decrease with increasing temperature because molecular vibrations disturb intermolecular coupling.

2.3 Field-Effect Transistors

Field-effect transistors (FETs) are three-terminal devices in which an electric field controls the conductivity of a semiconductor channel. The gate electrode is separated from the semiconductor by an insulating layer, so the gate voltage modulates the channel charge without ideally causing a steady gate current. This principle makes the FET a voltage-controlled device that can be used as a switch, a variable resistor, or an amplifier. In an organic field-effect transistor (OFET), the active channel is an organic semiconductor, and the mobile charge carriers are usually confined to a thin accumulation layer near the semiconductor–dielectric interface [2, 1].

For this thesis, the FET is important in two related ways. First, it provides the standard framework for describing gate-induced transport in organic thin films. Second, the same electrostatic control of a sheet conductance is used in the gated van der Pauw method discussed later. The following section therefore summarizes the operating regimes, the ideal current–voltage relations, and the small-signal concepts needed to interpret the measurements.

2.3.1 Operation of Field-Effect Transistors

A FET consists of a gate electrode, a gate dielectric, a semiconductor layer, and two contacts called source and drain. The source is used as the reference terminal. The gate–source voltage controls the density of mobile carriers in the channel, while the drain–source voltage creates the lateral electric field that drives these carriers. Thin-film transistors can be fabricated in several geometries, for example as bottom-gate or top-gate and bottom-contact or top-contact devices. These geometries affect injection and interface quality, but the underlying field-effect mechanism remains the same [2, 3].

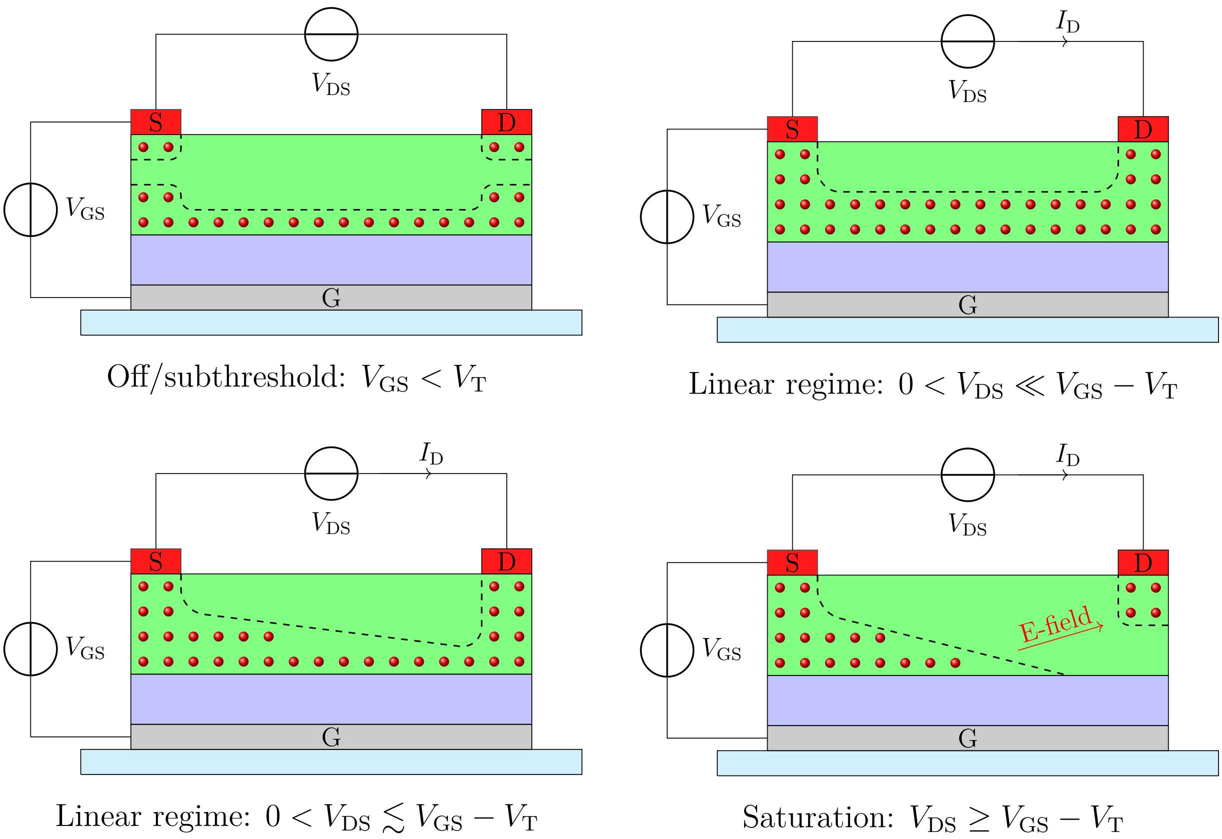

Figure 2.8: Schematic side views of the main operating regimes of a p-channel field-effect transistor. The voltage labels denote the positive bias magnitudes defined in Equation 2.34. Increasing gate bias forms a hole accumulation layer at the semiconductor–dielectric interface. Increasing drain bias makes the channel charge nonuniform and eventually produces pinch-off near the drain.

The devices considered in this work are operated as p-channel transistors. A gate voltage that is negative with respect to the source attracts holes to the dielectric interface. A negative drain voltage then drives the accumulated holes through the channel. To keep the notation compact, the following discussion uses positive magnitudes for the relevant terminal quantities,

$$V_{\mathrm{GS}} = |V_{\mathrm{G}} - V_{\mathrm{S}}|\,, \qquad V_{\mathrm{DS}} = |V_{\mathrm{D}} - V_{\mathrm{S}}|\,, \qquad I_{\mathrm{D}} = |I_{\mathrm{D,phys}}|\,. \tag{2.34}$$

The threshold voltage V_{T} is also written as a positive magnitude. With this convention, the mathematical form of the equations is the same as for an n-channel transistor, although the physical carriers are holes.

In the gradual-channel picture, the local channel potential is denoted by $V(x)$, with $V(0) = 0$ at the source and $V(L) = V_{\mathrm{DS}}$ at the drain. Above threshold, the accumulated sheet charge is approximated by

$$Q_{\mathrm{p}}(x) = C_{\mathrm{i}}\left[V_{\mathrm{GS}} - V_{\mathrm{T}} - V(x)\right], \tag{2.35}$$

provided that the expression in brackets is positive. Here C_{i} is the gate-insulator capacitance per unit area, and the two-dimensional hole density is $p_{\mathrm{2D}}(x) = Q_{\mathrm{p}}(x)/q$. This relation shows why the channel is not controlled by the gate voltage alone. At finite drain bias, the local potential also changes along the channel, so the charge density can be nearly uniform at small V_{DS} but strongly nonuniform at larger V_{DS} [26, 27].

The operating regimes in Figure 2.8 can then be summarized as follows. Below threshold, only a small mobile charge density is present and the device is nominally off. For $V_{\mathrm{GS}} > V_{\mathrm{T}}$ and small V_{DS}, a continuous accumulation channel connects source and drain, and the transistor behaves approximately as a gate-controlled resistor. When V_{DS} approaches $V_{\mathrm{GS}} - V_{\mathrm{T}}$, the channel charge near the drain is reduced. At $V_{\mathrm{DS}} \geq V_{\mathrm{GS}} - V_{\mathrm{T}}$, the ideal model predicts pinch-off at the drain side and the current enters the saturation regime.

The threshold voltage should not be understood as a sharp microscopic boundary between an empty and a conducting channel. It is an effective device parameter that includes the flat-band condition, localized states, fixed dielectric charge, mobile ions, and trapped charge. These contributions can shift V_{T} and can cause hysteresis between forward and backward gate-voltage sweeps. Such effects are especially relevant in OFETs, where the density of localized states is often broad [1, 2].

2.3.2 Current–Voltage Relationship

The standard current–voltage relations are obtained within the gradual-channel approximation. The model assumes a long and laterally uniform channel, a constant mobility μ, a uniform gate capacitance, negligible gate leakage, and no voltage loss at the source and drain contacts. The drift current at position x is written as

$$I_{\mathrm{D}} = W\mu Q_{\mathrm{p}}(x)\frac{\mathrm{d}V}{\mathrm{d}x}, \tag{2.36}$$

where W and L are the channel width and length. Substitution of Equation 2.35 and integration from source to drain gives

$$I_{\mathrm{D}} = \mu C_{\mathrm{i}}\frac{W}{L}\left[\left(V_{\mathrm{GS}} - V_{\mathrm{T}}\right)V_{\mathrm{DS}} - \frac{V_{\mathrm{DS}}^2}{2}\right], \qquad 0 \leq V_{\mathrm{DS}} < V_{\mathrm{GS}} - V_{\mathrm{T}}. \tag{2.37}$$

This expression describes the linear regime. At small drain bias, the quadratic term is negligible and

$$I_{\mathrm{D}} \approx \mu C_{\mathrm{i}} \frac{W}{L} \left(V_{\mathrm{GS}} - V_{\mathrm{T}} \right) V_{\mathrm{DS}}. \tag{2.38}$$

The channel conductance is then proportional to the gate overdrive, which is the basis for viewing the FET as a voltage-controlled resistor.

At the onset of pinch-off, $V_{\mathrm{DS}} = V_{\mathrm{GS}} - V_{\mathrm{T}}$. Inserting this condition into Equation 2.37 yields the ideal saturation current

$$I_{\mathrm{D,sat}} = \frac{1}{2} \mu C_{\mathrm{i}} \frac{W}{L} \left(V_{\mathrm{GS}} - V_{\mathrm{T}} \right)^2, \qquad V_{\mathrm{DS}} \geq V_{\mathrm{GS}} - V_{\mathrm{T}}. \tag{2.39}$$

In the ideal model, the saturation current is independent of drain voltage. Real output curves usually retain a finite slope because of channel-length modulation, field-dependent mobility, contact effects, or changes in trap occupation during the measurement. This nonideality is often represented phenomenologically by

$$I_{\mathrm{D,sat}} \approx \frac{1}{2} \mu C_{\mathrm{i}} \frac{W}{L} \left(V_{\mathrm{GS}} - V_{\mathrm{T}} \right)^2 \left(1 + \lambda V_{\mathrm{DS}} \right), \tag{2.40}$$

where λ is the channel-length-modulation parameter.

Experimentally, FETs are commonly described by transfer and output characteristics. A transfer curve records I_{D} as a function of V_{GS} at fixed drain bias and is used to estimate the threshold voltage, on/off ratio, subthreshold behavior, and apparent mobility. An output curve records I_{D} as a function of V_{DS} for different gate voltages and shows the transition from the linear regime to saturation. The same plots can also reveal injection limitations, for example nonlinear or S-shaped curves near the origin [3, 27].

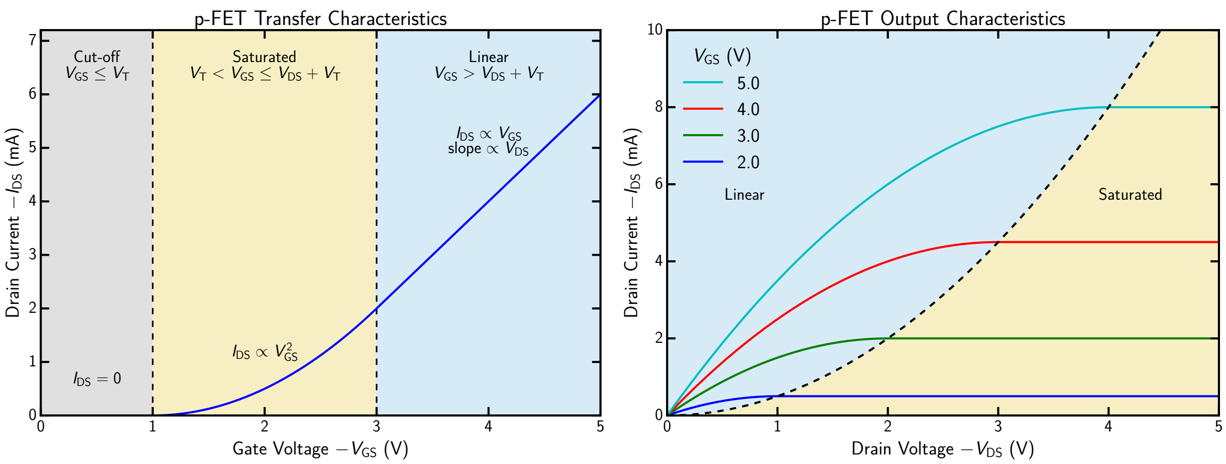

Figure 2.9: Idealized transfer characteristic at fixed drain bias (left) and output characteristics at several gate voltages (right). For consistency with Equation 2.34, the axes show the magnitudes of the p-channel voltages and current. In the linear regime, I_{D} is approximately linear in gate overdrive and drain voltage. In saturation, I_{D} is approximately quadratic in gate overdrive and ideally independent of drain voltage.

For a small and constant drain bias, Equation 2.38 gives the linear-regime mobility

$$\mu_{\mathrm{lin}} = \frac{L}{W C_{\mathrm{i}} V_{\mathrm{DS}}} \frac{\mathrm{d}I_{\mathrm{D}}}{\mathrm{d}V_{\mathrm{GS}}}. \tag{2.41}$$

In saturation, Equation 2.39 predicts a linear relation between $\sqrt{I_\mathrm{D}}$ and V_GS. The corresponding mobility estimate is

$$\mu_\mathrm{sat} = \frac{2L}{WC_\mathrm{i}} \left(\frac{\mathrm{d}\sqrt{I_\mathrm{D}}}{\mathrm{d}V_\mathrm{GS}} \right)^2 . \tag{2.42}$$

These values are model-dependent. If the mobility depends on carrier density, electric field, or contact injection, the extracted mobility is an apparent or differential quantity rather than a unique material constant.

Contact resistance is a major limitation in thin-film transistors. The measured terminal voltage is divided between the channel and the contacts,

$$R_\mathrm{tot} = R_\mathrm{C} + R_\mathrm{ch}. \tag{2.43}$$

A large or gate-dependent R_C changes the internal channel voltage and can distort the slope of the transfer curve. In OFETs, this can lead to mobility peaks or to a significant overestimation of the channel mobility [4, 3]. This is one reason why the contact-independent gated van der Pauw method, introduced in Section 2.4, is useful for the present work [5].

2.3.3 Small-Signal Amplification Properties

Besides switching, a FET can amplify small voltage variations. In a switching application, the transistor is driven between a high-resistance off state and a low-resistance on state. In an amplifier, by contrast, the device is biased at a stationary operating point and only small perturbations around this point are used. Figure 2.10 illustrates the switching principle for a simple resistive-load circuit.

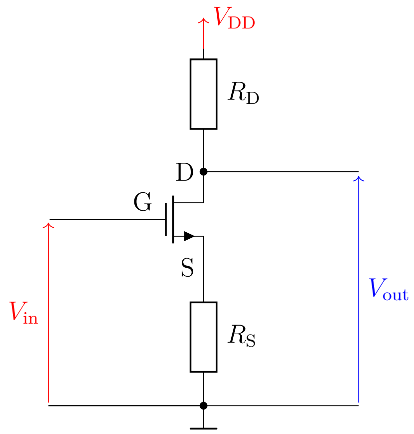

Figure 2.10: FET used as a voltage-controlled switch in a resistive circuit. A change of gate voltage changes the channel resistance and therefore the output voltage at the drain. The schematic uses the conventional n-channel polarity; a p-channel implementation requires reversed polarities.

At an operating point Q, the terminal quantities are written as a DC bias plus a small time-dependent signal,

$$V_\mathrm{GS}(t) = V_\mathrm{GS,Q} + v_\mathrm{gs}(t), \qquad V_\mathrm{DS}(t) = V_\mathrm{DS,Q} + v_\mathrm{ds}(t), \qquad I_\mathrm{D}(t) = I_\mathrm{D,Q} + i_\mathrm{d}(t). \tag{2.44}$$

Linearization of the current–voltage relation gives

$$i_\mathrm{d} = g_\mathrm{m} v_\mathrm{gs} + g_\mathrm{ds} v_\mathrm{ds}, \tag{2.45}$$

where

$$g_\mathrm{m} = \left.\frac{\partial I_\mathrm{D}}{\partial V_\mathrm{GS}}\right|_{V_\mathrm{DS}} \tag{2.46}$$

is the transconductance and

$$g_\mathrm{ds} = \left.\frac{\partial I_\mathrm{D}}{\partial V_\mathrm{DS}}\right|_{V_\mathrm{GS}}, \qquad r_\mathrm{o} = \frac{1}{g_\mathrm{ds}} \tag{2.47}$$

is the output conductance, with r_o the corresponding output resistance. The transconductance converts an input-voltage variation into a current variation, while the output resistance describes the finite slope of the output curve in saturation.

Using the ideal saturation expression, the transconductance becomes

$$g_\mathrm{m}(V_\mathrm{GS,Q}) = \mu C_\mathrm{i} \frac{W}{L} \left(V_\mathrm{GS,Q} - V_\mathrm{T}\right). \tag{2.48}$$

Thus, the small-signal response increases with mobility, gate capacitance, width-to-length ratio, and bias current. In the ideal saturation model, $g_\mathrm{ds} = 0$ and r_o is infinite. With channel-length modulation, $r_\mathrm{o} \approx 1/(\lambda I_\mathrm{D,Q})$.

Figure 2.11 shows a common-source amplifier. The resistors R_1 and R_2 set the gate bias, R_D converts drain-current variations into an output voltage, and R_S stabilizes the operating point. The coupling capacitors separate the DC bias from neighboring circuits. If the source-bypass capacitor acts as a short circuit for the signal frequency, the mid-band voltage gain is approximately

$$A_\mathrm{v} = \frac{v_\mathrm{out}}{v_\mathrm{in}} \approx -g_\mathrm{m} \left(R_\mathrm{D} \parallel r_\mathrm{o} \parallel R_\mathrm{L}\right). \tag{2.49}$$

The minus sign denotes phase inversion. If R_S is not bypassed, source degeneration reduces the gain approximately to

$$A_\mathrm{v} \approx -\frac{g_\mathrm{m} \left(R_\mathrm{D} \parallel r_\mathrm{o} \parallel R_\mathrm{L}\right)}{1 + g_\mathrm{m} R_\mathrm{S}}, \tag{2.50}$$

but improves bias stability and linearity. Because the gate is insulated, the low-frequency input resistance of an ideal FET is very high.

The useful bias point for voltage amplification lies where the large-signal transfer curve has a sufficiently steep and approximately linear slope, while the output voltage still has headroom for both positive and negative signal swings. This is illustrated in Figure 2.12. Biasing too close to cut-off gives a small transconductance, whereas biasing too far into the linear regime reduces the output resistance and gain. A large input signal moves the device over a curved part of the transfer characteristic and therefore causes nonlinear distortion.

At higher frequencies, the gain is reduced by the gate–source and gate–drain capacitances, wiring capacitances, contact resistance, and slow charging or trapping processes in the semiconductor. A useful estimate of the intrinsic current-gain bandwidth is the transition frequency

$$f_\mathrm{T} \approx \frac{g_\mathrm{m}}{2\pi \left(C_\mathrm{GS} + C_\mathrm{GD}\right)}. \tag{2.51}$$

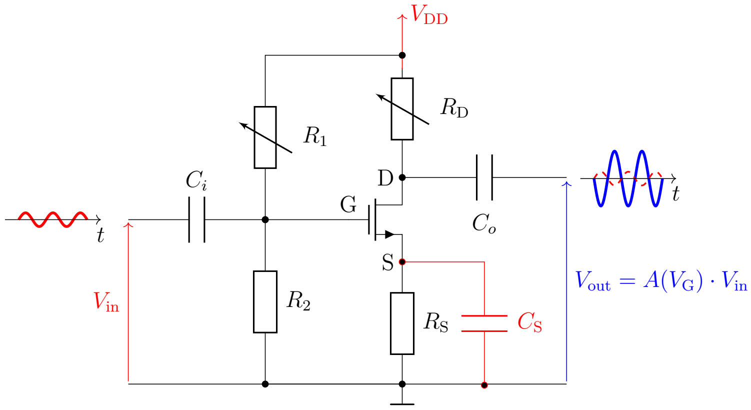

Figure 2.11: Common-source FET amplifier with a resistive drain load, gate-bias network, source stabilization, and coupling capacitors. The source-bypass capacitor increases the AC gain by suppressing source degeneration over its effective frequency range. The output signal is amplified and inverted relative to the input signal.

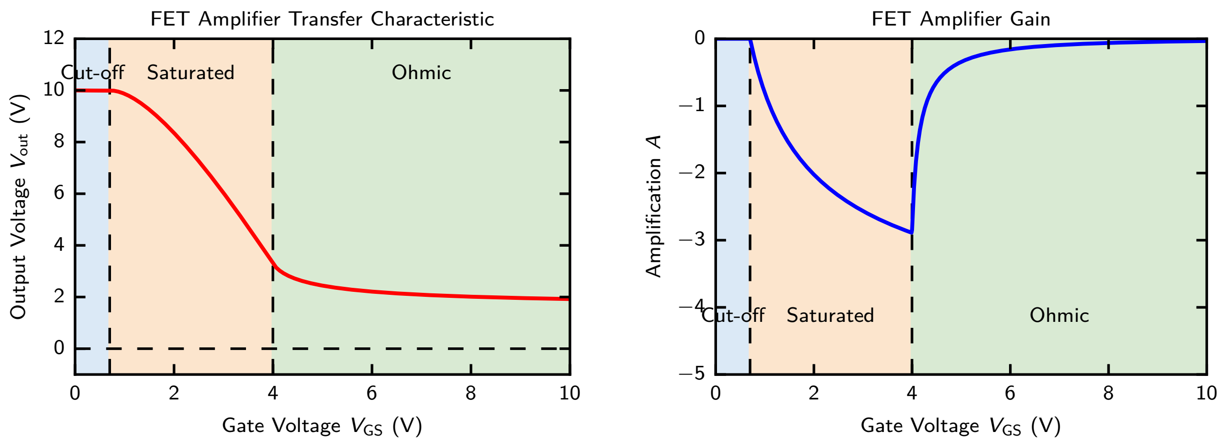

Figure 2.12: Example of a resistively loaded FET amplifier. The left panel shows the static output voltage as a function of gate bias. The local slope of this curve determines the voltage gain shown on the right. The gain is negative in the inverting common-source configuration and is largest in magnitude within the transition region between cut-off and strongly ohmic operation.

For OFETs, the experimentally observed bandwidth can be considerably lower than this estimate, because contacts, distributed channel resistance, and trapping introduce additional time constants [2, 3]. The DC characteristics and the dynamic amplifier response therefore provide complementary information about charge accumulation, transport, and trapping in the investigated devices.

2.4 Contact-Independent Gated Van-der-Pauw Method

The mobility extracted from a conventional two-terminal field-effect transistor is obtained from the gate dependence of the terminal current. This procedure is reliable only when the voltage drop at the source and drain contacts is negligible compared with the voltage drop across the accumulated channel. In organic thin-film transistors, however, the contact resistance can be large and strongly gate dependent. It can therefore change the slope of a transfer curve and produce an apparent mobility that is either underestimated or, in the presence of gate-modulated injection, strongly overestimated [4, 3].

The gated van der Pauw (gVDP) method avoids this ambiguity by combining electrostatic charge accumulation with a four-terminal measurement of the sheet conductance. Rolin et al. introduced the method for organic semiconductors and demonstrated that it yields the long-range mobility of the accumulation layer independently of the voltage losses at the current-injecting contacts [5]. More recent work by Lee et al. showed that the same principle corrects contact-induced mobility overestimation in short-channel thin-film transistors and can be extended to separate channel and contact resistances within one device [7, 8]. The present thesis uses the current-driven form of the method, hereafter denoted C-gVDP.

2.4.1 Classical Van-der-Pauw Method

The classical van der Pauw method determines the sheet resistance of a thin conducting film without assigning a rectangular channel length and width. Its standard assumptions are that the film is homogeneous, approximately isotropic in the plane, of uniform thickness, simply connected, and contacted by sufficiently small electrodes located at its boundary. Under these conditions, the sheet resistance is determined by the potential distribution inside the film rather than by its external shape.

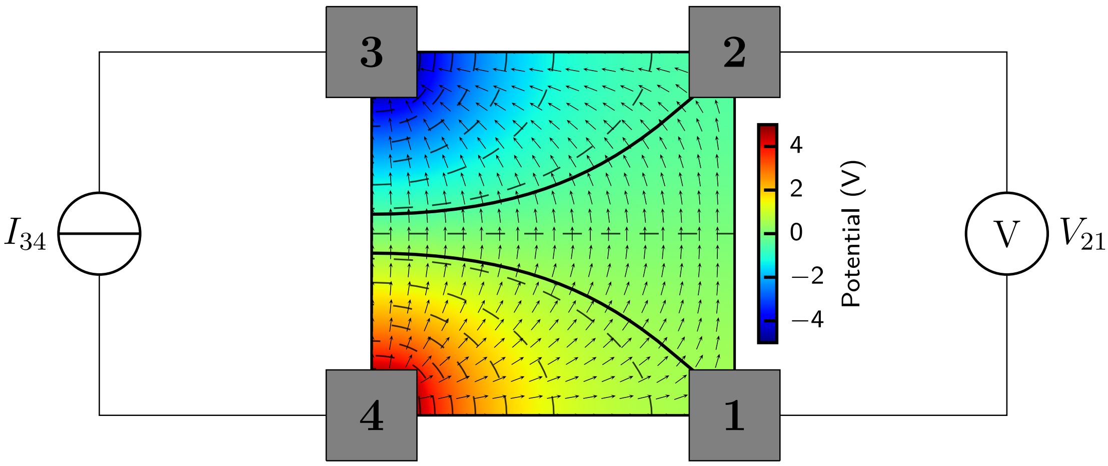

Figure 2.13: Classical van der Pauw geometry. A current I_{34} is driven between contacts 3 and 4, while the voltage V_{21} is measured between the remaining high-impedance voltage probes. The dashed contours indicate equipotential lines; in particular, the contours passing through contacts 1 and 2 mark the two potentials whose difference is measured as V_{21}. The quiver arrows show the local electric-field direction and, in the Drude model, the corresponding current-density direction. Adapted from Ref. [5].

For the configuration in Figure 2.13, the measured four-terminal resistance is

$$R_{34,21} = \left|\frac{V_{21}}{I_{34}}\right|. \tag{2.52}$$

A second independent resistance is obtained after rotating the terminals, for example $R_{23,14} = |V_{14}/I_{23}|$. The sheet resistance R_s is the solution of the van der Pauw equation

$$\exp\left(-\frac{\pi R_{34,21}}{R_s}\right) + \exp\left(-\frac{\pi R_{23,14}}{R_s}\right) = 1. \tag{2.53}$$

For a fourfold-symmetric square or clover-leaf device, the complementary resistances are ideally equal. With their averaged value $\overline{R}$, the sheet resistance and sheet conductance reduce to

$$R_s = \frac{\pi}{\ln 2}\overline{R}, \qquad \sigma_s = \frac{1}{R_s} = \frac{\ln 2}{\pi\overline{R}}. \tag{2.54}$$

In practice, the current direction is reversed and all four sides are measured, which gives eight resistance values. Their average suppresses offsets caused by small geometrical asymmetries, contact misalignment, and thermoelectric voltages [5, 7].

The voltage probes ideally draw no current. Consequently, their metal– semiconductor contact resistances do not produce a series voltage drop in the measured transverse voltage. Contact resistance may still affect the voltage required to inject the prescribed current, but it does not directly enter Equation (2.54).

2.4.2 Current-Driven Gated Van-der-Pauw Geometry

The gated method adds a common gate electrode beneath the dielectric and the semiconductor film. In the contact notation used in this thesis, contact 4 is the electrical reference, a constant current I_{34} is driven between contacts 3 and 4, and contacts 1 and 2 are used only as voltage probes. The measured probe voltages are V_{14} and V_{24}, from which the transverse van der Pauw voltage and the mean potential of the probed region are defined as

$$V_{12} = V_{14} - V_{24}, \qquad V_C = \frac{V_{14} + V_{24}}{2}. \tag{2.55}$$

The corresponding four-terminal resistance is

$$R_{34,12} = \left|\frac{V_{12}}{I_{34}}\right|. \tag{2.56}$$

After reversing and rotating the contact configuration, the averaged resistance $\overline{R}$ is inserted into Equation (2.54) to obtain the intrinsic sheet conductance.

It is important to distinguish the two measured voltage drops. The longitudinal voltage V_{34} is the voltage that the current source must apply to establish I_{34} and therefore contains voltage losses in both the channel and the injecting contacts. In contrast, V_{12} is sensed by passive probes inside the film and is used to determine σ_s. Lee et al. refer to these as the contact-sensitive transfer regime and the contact-independent conductance regime, respectively [8]. The current-driven method used here employs the conductance regime for mobility extraction.

Figure 2.14 illustrates why the internal potential V_C must be included in the analysis. The gate does not act relative to an everywhere-grounded semiconductor sheet; it acts relative

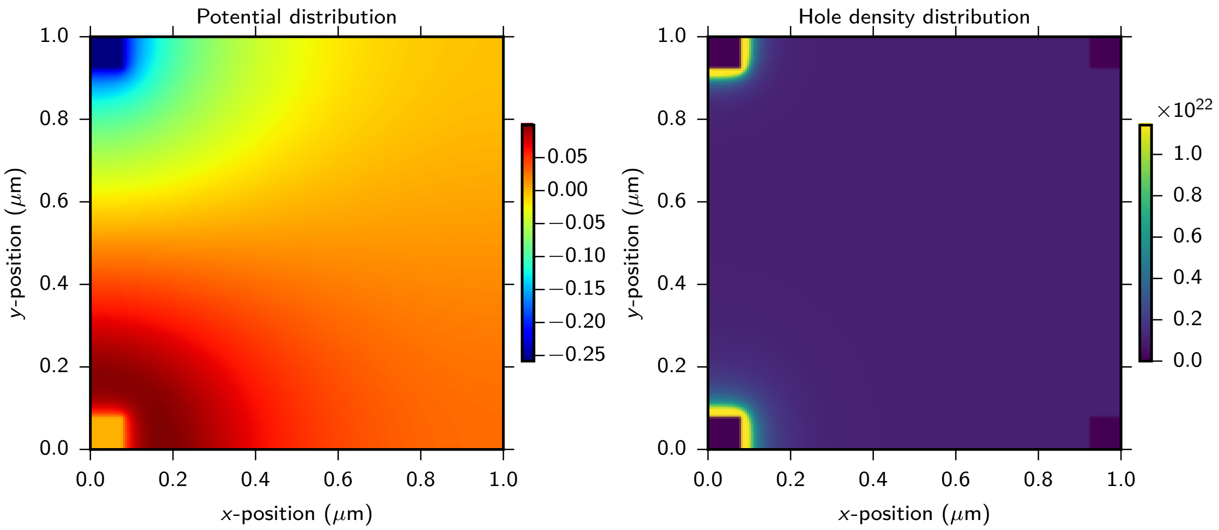

Figure 2.14: Numerically calculated potential and hole-density distributions in a gated van der Pauw structure. A finite measurement current produces a spatially varying sheet potential. Since the local gate-induced charge depends on the voltage difference between the gate and the semiconductor, the carrier density is spatially varying as well. The two non-injecting probe potentials define the representative channel potential V_C used in the charge-density correction. The hole-density map shows pronounced accumulation of holes near the current-injecting contacts, where the imposed current enters and leaves the organic film. This local enhancement can result from the electrical boundary conditions and does not, by itself, demonstrate the presence of a Schottky barrier.

to the local semiconductor potential. Even when contact 4 is grounded, the central region of the film acquires a finite potential under current flow. Neglecting this potential is equivalent to assuming $V_\mathrm{C} = 0$, which is justified only in the limit of a vanishing measurement current or a negligible voltage drop inside the film.

2.4.3 Gate-Induced Sheet Density and Mobility Extraction

Within the gradual-channel and parallel-plate-capacitor approximations, the local gate-induced sheet charge at a semiconductor potential ϕ is

$$Q_\mathrm{2D}(\phi) = C_\mathrm{i}\left(V_\mathrm{G4} - \phi - V_\mathrm{T}\right), \tag{2.57}$$

where C_i is the gate-insulator capacitance per unit area, V_G4 is the gate voltage relative to contact 4, and V_T is the threshold voltage in the same voltage convention. Approximating the potential of the region sampled by the voltage probes by $\phi \simeq V_\mathrm{C}$ gives the modified two-dimensional carrier density used in this thesis,

$$\boxed{n_\mathrm{2D} = \frac{C_\mathrm{i}}{q}\left(V_\mathrm{G4} - V_\mathrm{C} - V_\mathrm{T}\right)}. \tag{2.58}$$

Equation (2.58) is a signed carrier density. For the p-type devices investigated here, the quantity in parentheses is negative when holes are accumulated. The positive hole number density is therefore

$$p_\mathrm{2D} = \frac{C_\mathrm{i}}{q}\left|V_\mathrm{G4} - V_\mathrm{C} - V_\mathrm{T}\right|. \tag{2.59}$$

The sheet conductance is related to sheet density and mobility by

$$\sigma_{\mathrm{s}} = q p_{\mathrm{2D}} \mu_{\mathrm{gVDP}}. \tag{2.60}$$

Combining Equations (2.59) and (2.60) yields

$$\sigma_{\mathrm{s}} = \mu_{\mathrm{gVDP}} C_{\mathrm{i}} \left| V_{\mathrm{G4}} - V_{\mathrm{C}} - V_{\mathrm{T}} \right| . \tag{2.61}$$

A derivation from the gradual-channel drift equation is provided in Appendix B.

If the mobility is approximately constant over the fitted high-charge-density range, a plot of σ_{s} against the internal gate voltage $V_{\mathrm{G4}} - V_{\mathrm{C}}$ is linear. The mobility follows from its slope,

$$\mu_{\mathrm{gVDP}} = \frac{1}{C_{\mathrm{i}}} \left| \frac{\mathrm{d}\sigma_{\mathrm{s}}}{\mathrm{d}\left(V_{\mathrm{G4}} - V_{\mathrm{C}}\right)} \right| , \tag{2.62}$$

and the horizontal intercept gives

$$V_{\mathrm{G4}} - V_{\mathrm{C}} = V_{\mathrm{T}}. \tag{2.63}$$

The correction by V_{C} is essential. Using only $V_{\mathrm{G4}} - V_{\mathrm{T}}$ assigns the full gate voltage to the dielectric, even though part of it is offset by the finite potential of the conducting sheet. This can introduce a current-dependent error in both the carrier density and the extracted threshold voltage.

Equation (2.62) should not be interpreted as a universal pointwise mobility formula when μ itself changes strongly with carrier density. In that case, the slope contains both the mobility and its gate-voltage dependence. The constant-mobility value is therefore extracted only from a region in which σ_{s} is well described by a straight line, as proposed by Rolin et al. [5].

2.4.4 Operating Regimes and Measurement Validity

In a C-gVDP measurement, the source meter continuously adjusts V_{34} to keep I_{34} constant while the gate voltage is swept. Close to the off-state, the film resistance and the required injection voltage can become large. The device then enters a saturation-like regime in which the injecting-contact potential follows the gate voltage and the current source may approach its compliance limit. At sufficiently strong accumulation, the required longitudinal voltage becomes small and the probed region behaves as a linear, approximately homogeneous conducting sheet. Rolin et al. identified these regimes through the measured contact and probe potentials and extracted the mobility from the high-charge-density linear regime [5].

Reliable gVDP extraction is therefore based on the following experimental checks:

- the current source remains below its voltage-compliance limit and the gate leakage remains negligible;
- V_{12} scales linearly with I_{34}, so that the measurement is in the small-signal or ohmic transport regime;
- conductance curves measured at different current levels collapse when plotted against $V_{\mathrm{G4}} - V_{\mathrm{C}}$;
- mobility and threshold voltage are fitted only over the linear high-charge-density part of the conductance curve.

The use of several current levels is more than a reproducibility check. Since V_{C} changes with the imposed current, agreement of the corrected conductance curves demonstrates that the measured sheet conductance is a property of the accumulated film rather than being dependent on the selected source current. This current independence is one of the central validity criteria used in the analysis in Chapter 4.

2.4.5 Meaning and Limitations of Contact Independence

The term *contact independent* refers specifically to the extraction of the sheet voltage from non-injecting, high-impedance probes. The contact voltage drops at the current electrodes are excluded from V_{12} and therefore do not appear directly in R_{s}. This is the essential distinction from a two-terminal OFET transfer curve, in which the measured voltage includes both channel and contact contributions.

The contacts are nevertheless not physically irrelevant. The injecting electrodes must still supply the prescribed current, and poor injection can produce large compliance voltages, nonlinear response, self-heating, or an insufficiently accumulated region. Furthermore, the van der Pauw analysis can be degraded by large contacts, disconnected regions, strong lateral inhomogeneity, anisotropic transport, substantial gate leakage, or a voltage probe with insufficient input impedance. Charge trapping and hysteresis may also shift V_{T} between sweep directions even when the sheet conductance itself is measured without a contact voltage drop.

Finally, μ_{gVDP} is the macroscopic, contact-independent mobility of carriers moving through the field-induced accumulation layer over the scale of the thin film. It includes long-range transport barriers such as grain boundaries, structural disorder, and trap-rich regions. It should therefore not be identified automatically with the microscopic intragrain mobility of an ideal crystal [5]. For the present thesis, this macroscopic accumulation-layer mobility is the relevant quantity because it characterizes the transport that governs actual thin-film transistor operation.

CHAPTER 3

DEVICE ARCHITECTURE

This chapter describes the architecture and fabrication route of the organic thin-film devices used for gated van der Pauw (gVDP) measurements. The devices investigated in this thesis were fabricated by other members of the research group; the process is summarized here to document the sample structure and to provide the necessary context for the electrical characterization. The layout combines a gate stack for controlling the charge density in the organic semiconductor with four spatially separated contacts for measuring the sheet conductance, and follows established procedures for organic thin-film devices used in the group.

3.1 Device Architecture

Layer	Function	Material and thickness
Layer 1	Gate contact	8 nm Cr
Layer 2	Insulator	300 nm PaN + 80 nm PS
Layer 3	Semiconductor	40 nm C8-BTBT
Layer 4	Injection contact	3 nm MoO_x
	Injection contact	20 nm Au
Layer 5	Contact pads	3 nm Cr
	Contact pads	150 nm Ag
	Contact pads	20 nm Au

Table 3.1: Layer sequence of the fabricated device, including material thicknesses and lithography masks. Adapted from Ref. [28].

The fabricated samples are bottom-gate thin-film devices designed for gated van der Pauw (gVDP) measurements. Their overall contact layout is shown in Figure 3.1, while Figure 3.2 gives a simplified schematic of the assembled layer and contact structure. A chromium layer forms the bottom gate and is covered by the insulating parylene-N (PaN) and polystyrene (PS) layers. The molecular semiconductor is deposited on top of this dielectric stack and forms the active layer in which field-effect-induced charge carriers move. Four metal contacts around the semiconductor serve as current-injection and voltage-probe contacts during the gVDP measurement, while the larger contact pads provide mechanically stable, low-resistance access for external wiring [5, 7].

A summary of the layer sequence is given in Table 3.1. The nominal thicknesses are target

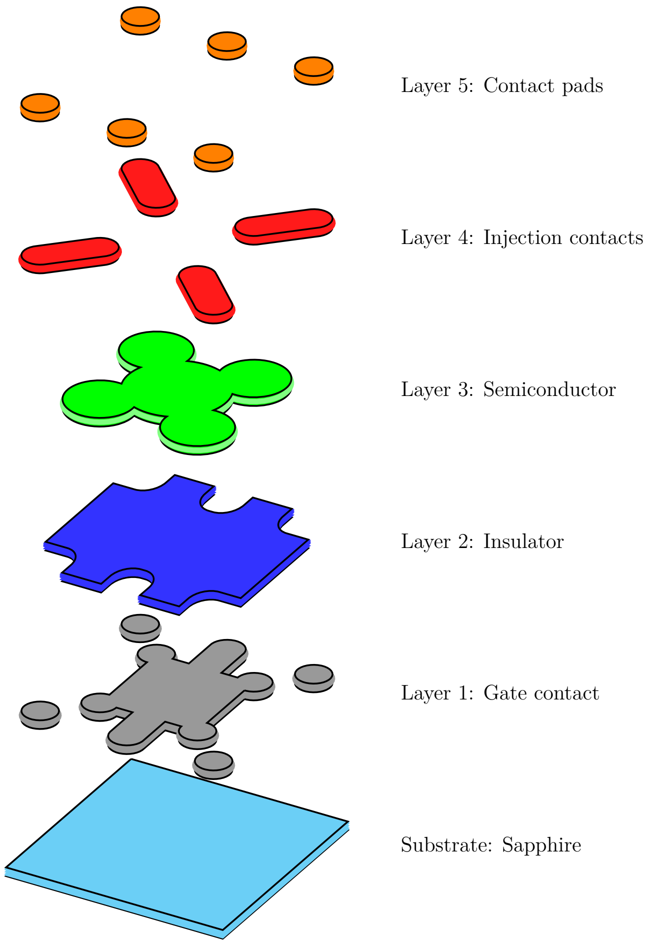

Figure 3.1: Device layout used for electrical characterization. The structure contains a bottom gate, a dielectric stack, a molecular semiconductor layer, four injection contacts, and larger contact pads for external wiring. Adapted from Ref. [28].

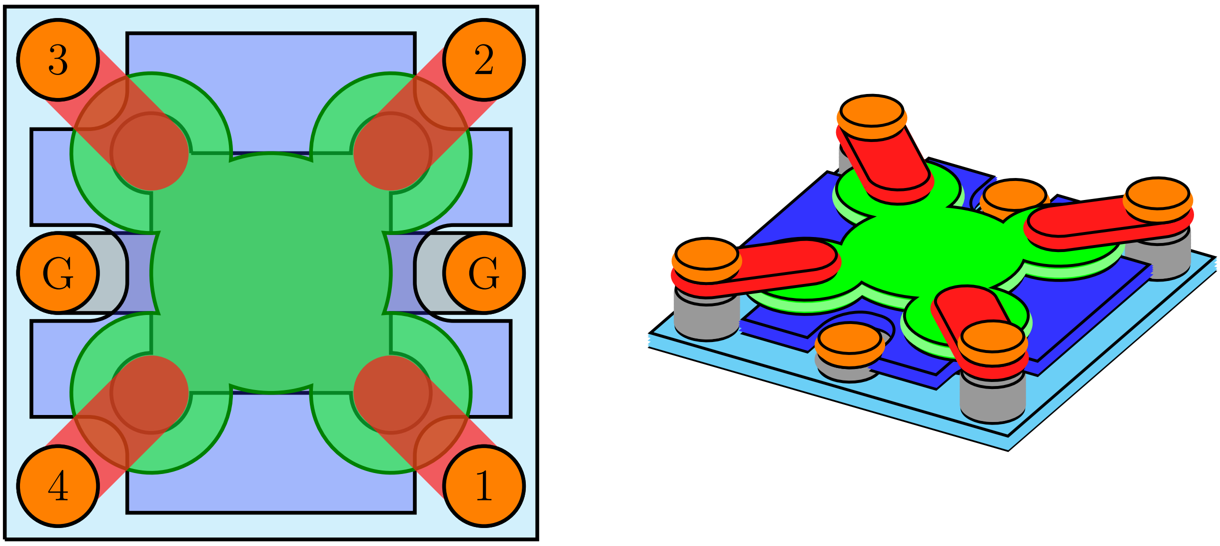

Figure 3.2: Schematic representation of the assembled OFET structure. The left panel shows the top-view contact layout, while the right panel shows the corresponding 3D layer structure. The central green region denotes the molecular semiconductor, the red diagonal paths indicate the four injection contacts, the orange pads provide external electrical access, and the side pads marked G connect to the bottom gate. Adapted from Ref. [28].

values; small deviations can occur due to deposition-rate variations, mask positioning, and the surface quality of the underlying layer.

The bottom chromium layer is used as the gate electrode. Chromium adheres well to the substrate and can be deposited as a thin, continuous film by physical vapor deposition. Its role is not to inject charge carriers, but to create an electric field across the dielectric stack.

The dielectric stack consists of PaN and PS. PaN provides the main electrical insulation between the gate and the semiconductor, while PS acts as a planarization layer. A smooth dielectric surface supports more reproducible molecular ordering, grain formation, and electrical measurements [2, 15].

The molecular semiconductor layer is the active layer of the device and must be continuous between the four contacts. The clover-leaf geometry provides a well-defined four-contact structure while still allowing relatively large metal contacts. The central region is mainly probed during the van der Pauw measurement, and the gate voltage changes the density of accumulated charge carriers close to the dielectric/semiconductor interface.

The injection contacts consist of MoO_x and Au. Au is stable and has a high work function, making it suitable for hole injection into many p-type organic semiconductors. The thin MoO_x interlayer improves the energetic alignment between the metal and the semiconductor and can lower the hole-injection barrier [16, 17, 3]. The contacts must overlap the semiconductor sufficiently without shorting neighboring contacts.

The final contact pads consist of Cr, Ag, and Au. Cr acts as an adhesion layer, Ag provides a thick and highly conductive pad, and the Au capping layer protects the surface during probing or wiring. These pads are larger than the injection contacts because they are used for external electrical access and do not define the active transport region.

The architecture is well suited for gVDP measurements because voltage is sensed at contacts that ideally carry no current. This reduces the influence of contact resistance on the extracted sheet conductance, unlike in conventional two-terminal or transistor measurements where contact resistance can distort the apparent mobility and threshold voltage [5, 4, 3].

3.2 Fabrication Methods

Fabrication follows a layer-by-layer process in which each functional layer is deposited after the previous layer has been completed and inspected. Shadow masks define the metallic and semiconducting layers, whereas the dielectric layers cover the full device area. The main fabrication steps are physical vapor deposition (PVD), chemical vapor deposition (CVD), and spin coating.

3.2.1 Physical Vapor Deposition

Physical vapor deposition is used for the metallic layers and for the molecular semiconductor. The samples and shadow mask are mounted in the substrate holder, while the source material is placed in the lower part of the vacuum chamber. Metals are heated until they evaporate, and the organic semiconductor is typically evaporated from a Knudsen cell. The evaporated material then travels through the chamber and condenses on the sample surface.

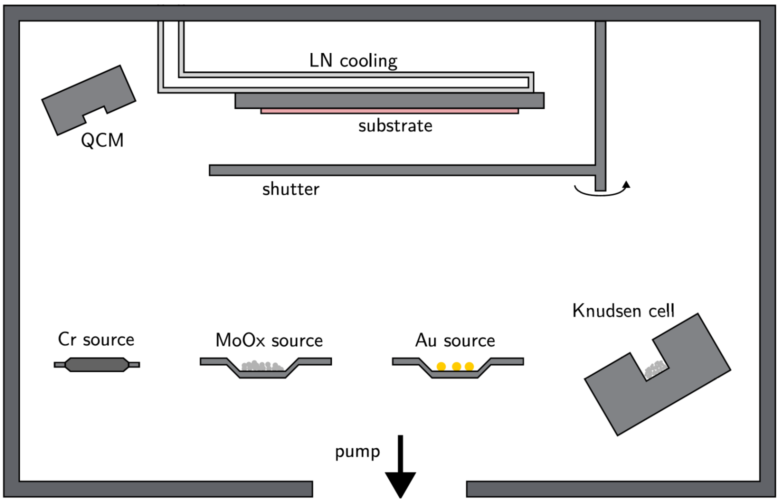

Figure 3.3: Schematic of the physical vapor deposition (PVD) system. Figure from Ref. [21].

Deposition is performed under high-vacuum conditions to reduce contamination by residual gas molecules and to make the particle mean free path larger than the chamber dimensions. The particles can therefore travel approximately ballistically from the source to the substrate, which is important because shadow-mask patterns are defined by the line-of-sight path through the mask openings.

A shutter between source and sample remains closed while the source is heated and possible impurities are removed. Once the deposition rate has stabilized, the shutter is opened and film growth begins. The film thickness is monitored with a quartz-crystal microbalance, and deposition is stopped when the desired thickness is reached.

In the fabricated devices, PVD is used for the Cr gate, the molecular semiconductor, the MoO_x/Au injection contacts, and the Cr/Ag/Au contact pads. Careful mask alignment is essential because the injection contacts must overlap the semiconductor while remaining electrically separated from one another.

3.2.2 Chemical Vapor Deposition of Parylene-N

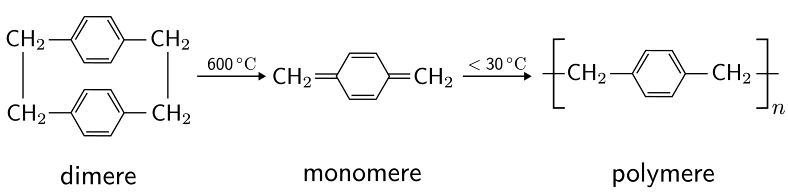

Figure 3.4: Schematic of the chemical synthesis of parylene-N (PaN). Figure from Ref. [21].

Parylene-N is deposited by chemical vapor deposition. The parylene precursor is first evaporated and then converted into reactive monomers at high temperature. These monomers are transported into the deposition chamber, where they polymerize on the sample surface and form a conformal insulating film.

In the device stack, PaN is the main gate dielectric. It separates the gate electrode from the

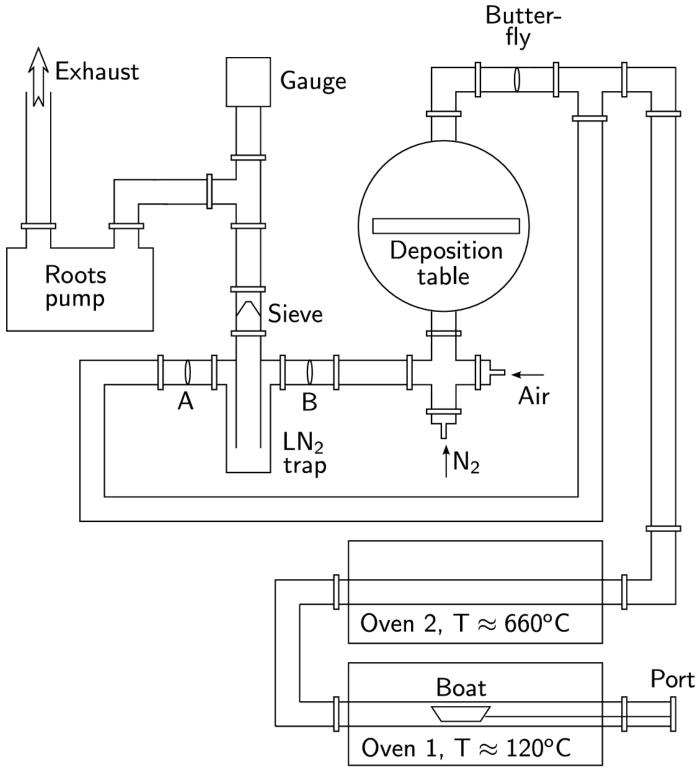

Figure 3.5: Schematic of the chemical vapor deposition (CVD) system used for PaN deposition. Figure from Ref. [28].

semiconductor and prevents direct leakage through the gate stack. Gas-phase deposition allows PaN to cover the gate structure uniformly, reducing the risk of pinholes, local thickness variations, leakage, and electrical breakdown.

Before deposition, the system is evacuated and the sample holder can be heated to remove adsorbed water. During deposition, the sample temperature is kept low enough for polymerization on the surface. The target PaN thickness is 300 nm. If the dielectric covers the gate contact areas, they are opened locally before the final contacting step.

3.2.3 Spin Coating of the PS Planarization Layer

The PS layer is deposited by spin coating. A polymer solution is dispensed onto the sample, which is then rotated at high speed. Centrifugal forces spread the solution over the surface while solvent evaporates, leaving a thin polymer film. The final thickness depends on the polymer concentration, solvent, spin speed, spin time, and drying conditions.

In this device architecture, PS planarizes the PaN surface and provides a more homogeneous interface for growth of the molecular semiconductor. This is important because molecular ordering and grain formation are sensitive to the dielectric surface. A smoother and chemically more homogeneous surface can improve film reproducibility [2, 15].

The target PS thickness is 80 nm. After spin coating, the film is dried to remove residual solvent. The resulting PaN/PS dielectric stack must be continuous and electrically insulating because defects can cause gate leakage and make field-effect measurements unreliable.

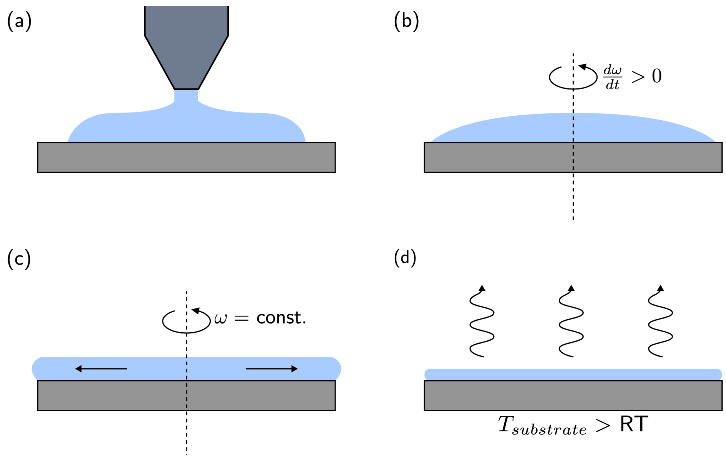

Figure 3.6: Schematic of the spin coating procedure. Figure from Ref. [21].

3.3 Images of Fabricated Devices

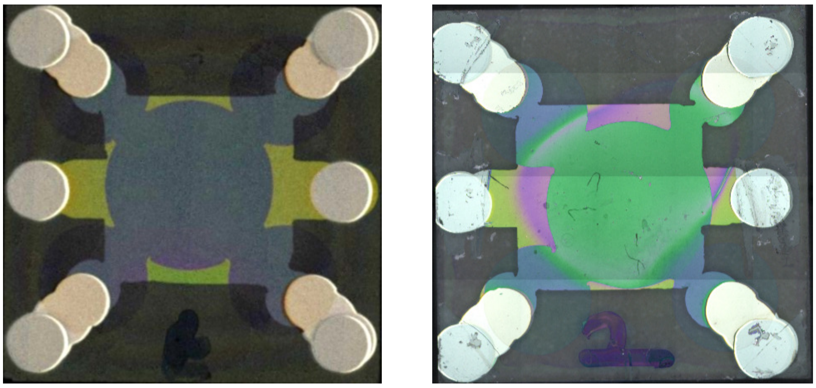

Figure 3.7: Fabricated devices under the microscope. Left: optically well-defined device. Right: device with visible defects. Photos from internal laboratory archive.

After fabrication, the devices were inspected under an optical microscope because their electrical behavior depends strongly on the integrity of the active film and the separation of the four contacts [2, 5]. A device was considered suitable for measurement when the semiconductor layer was continuous, the injection contacts were clearly separated, and the contact pads could be reached reliably with the probes.

Figure 3.7 shows an optically well-defined device and a defective device. In a well-defined device, the layers are aligned so that the four contacts overlap the semiconductor without shorting neighboring contacts, and the central semiconductor region is continuous. Defective devices can show interrupted metal paths, incomplete semiconductor coverage, mask misalignment, dust particles, or unwanted metal bridges, which may cause open contacts, short circuits, leakage currents, or non-reproducible results.

Optical inspection was followed by electrical checks. The resistance between contact pads was measured to identify open and short circuits, and gate leakage was checked before large gate voltages were applied. Only devices that passed these tests were used for the gVDP measurements described in the following chapters [5, 7].

CHAPTER 4

EXPERIMENTAL METHODS AND RESULTS

4.1 Current-Voltage Characteristics

Current-voltage (I–V) characteristics provide an initial assessment of the electrical quality and field-effect behaviour of the fabricated organic field-effect transistors (OFETs). The measurement configuration is shown schematically in Figure 4.1. The voltage between contacts 3 and 4 is denoted by V_{34}, and the corresponding current is denoted by I_{34}. The gate voltage is applied between the gate electrode and contact 4 and is denoted by V_{G4}, while the gate leakage current is denoted by I_{G4}.

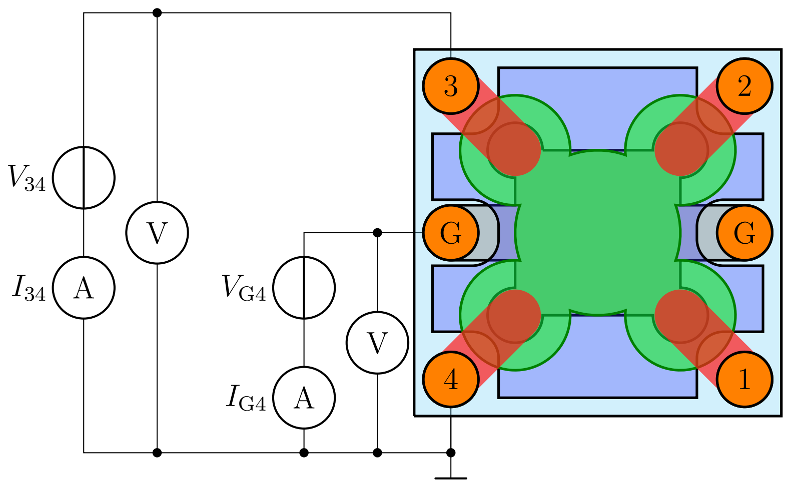

Figure 4.1: Circuit used for the current-voltage characterization of the OFET. The source–drain branch measures V_{34} and I_{34} between contacts 3 and 4, while the gate branch applies V_{G4} and monitors the gate leakage current I_{G4}. Adapted from Ref. [28].

Because the devices operate as p-type channels, negative gate voltages accumulate holes at the semiconductor–dielectric interface [2, 12]. The measured data are therefore plotted using the sign convention $-V_{G4}$, $-V_{34}$, and $-I_{34}$, so that increasing positive values on the axes correspond to stronger negative bias during the experiment. This convention also makes the transfer and output curves easier to compare, because larger plotted values indicate stronger hole accumulation or a larger hole current.

The two most important representations are the transfer and output characteristics. The transfer characteristic shows how efficiently the gate voltage controls the channel current at fixed drain voltage, whereas the output characteristic shows how the current responds to the applied drain voltage at fixed gate voltage. Both measurements include contributions from the semiconductor channel, charge injection at the contacts, trap states, and possible bias-stress or hysteresis effects. Mobility values extracted directly from two-terminal OFET curves must therefore be interpreted as apparent mobilities, especially when contact resistance is not negligible [3, 4]. This limitation motivates the use of contact-independent methods, such as the gated van der Pauw method discussed in the following sections [5].

The gate leakage current was monitored as an additional quality criterion using the gate branch shown in Figure 4.1. This measurement verifies the integrity of the gate insulator and confirms that the measured source–drain current is not dominated by unintended current through the dielectric [19].

4.1.1 Insulator Leakage Current

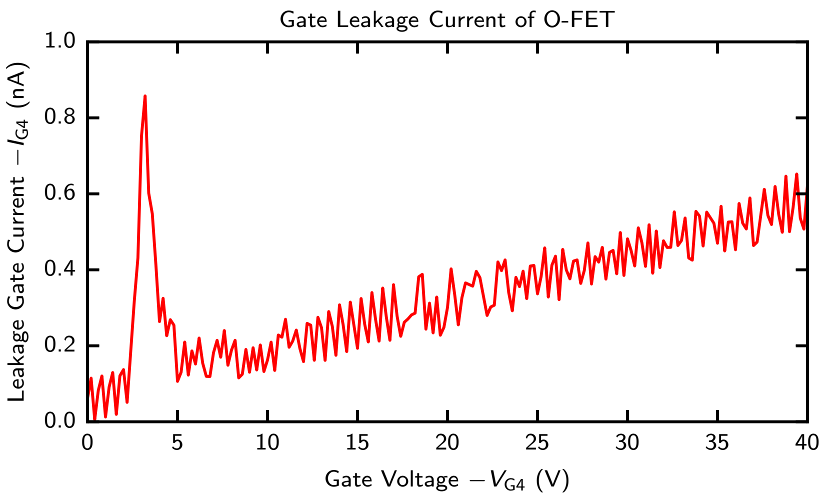

Figure 4.2: Gate-insulator leakage current of the OFET, measured as $-I_{G4}$ while sweeping the gate voltage $-V_{G4}$. The leakage current remains in the nanoampere range over the full voltage sweep, well below the 1 µA limit. The isolated peak at low gate voltage is interpreted as a transient caused by a small dust particle, which is subsequently swept out by the applied electric field.

Figure 4.2 shows the gate leakage current measured during the gate-voltage sweep. Apart from one pronounced transient peak at low gate voltage, the leakage remains below approximately 1 nA over the investigated voltage range. This value is several orders of magnitude below the limit of 1 µA, indicating that the gate dielectric is electrically insulating under the measurement conditions.

The narrow peak is not followed by a sustained increase of leakage current. It is therefore most likely caused by a local, non-permanent defect such as a dust particle or surface residue near the gate-insulator interface. Once the electric field is increased, this contamination-related conduction path is swept out and the leakage current returns to the low background level. The absence of a persistent high-current state indicates that the feature does not correspond to dielectric breakdown. The leakage measurement therefore supports the validity of the subsequent transfer and output characteristics, because the measured drain current is dominated by channel transport rather than by parasitic gate leakage.

4.1.2 Transfer Characteristic

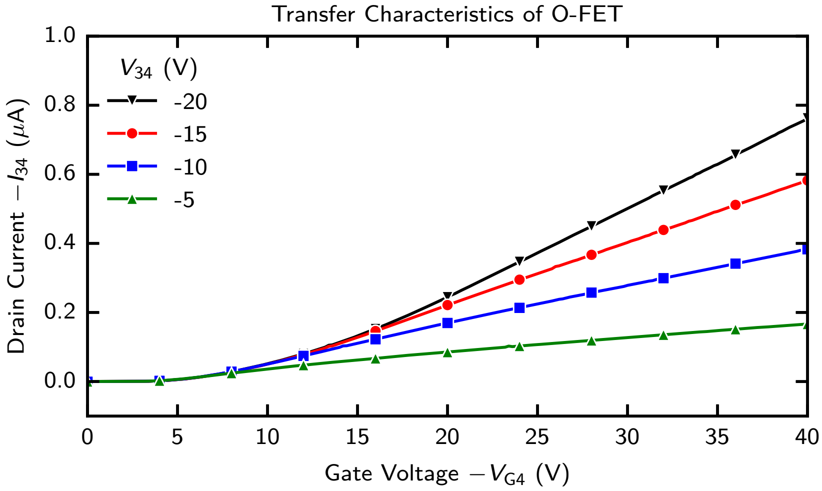

Figure 4.3: Transfer characteristic of the organic FET, measured as drain current $-I_{34}$ as a function of gate voltage $-V_{\mathrm{G4}}$ for different fixed drain voltages V_{34}.

Figure 4.3 shows the drain current as a function of gate voltage at fixed drain voltages. In this measurement, the gate voltage in Figure 4.1 is swept while the voltage V_{34} between contacts 3 and 4 is kept constant. At low $-V_{\mathrm{G4}}$, the device is close to the off-state, because only a small density of holes is accumulated at the semiconductor–dielectric interface. As $-V_{\mathrm{G4}}$ increases, the accumulated hole density rises and the channel becomes more conductive. The drain current therefore increases with gate-voltage overdrive, and the largest current is obtained at the largest drain bias, $V_{34} = -20\,\mathrm{V}$.

The transfer curves do not show an abrupt ideal turn-on. Instead, the current increases gradually with gate voltage. This behaviour is typical of organic thin-film devices and can arise from trap filling, energetic disorder, interface states, and contact-related voltage drops [22, 2, 3]. A threshold voltage can, in principle, be estimated by extrapolating the approximately linear high-current region of the transfer curve. However, the resulting value should be treated with care, because the measured two-terminal current is not determined solely by the intrinsic channel. Contact resistance and charge injection can change the apparent slope of the transfer curve and therefore affect both the extracted mobility and the threshold voltage [4, 3].

The same behaviour is shown in resistance form in Figure 4.4. The drain–source resistance is high near the off-state, where the accumulation layer is weak, and decreases strongly as the gate voltage becomes more negative. This decrease confirms that the gate voltage forms a progressively more conductive p-type channel between contacts 3 and 4.

The fit in Figure 4.4 does not extrapolate to zero resistance at high gate bias. Instead, it gives a finite background resistance. This offset indicates that the measured two-terminal resistance contains more than the gate-dependent channel resistance. A major contribution is expected from the metal–semiconductor contacts, where charge injection across a Schottky barrier adds a series resistance [3, 29]. The resistance analysis therefore supports the interpretation of the transfer curve: the device shows clear field-effect behaviour, but quantitative parameters extracted from the two-terminal measurement include contact and injection effects as well as the intrinsic channel response.

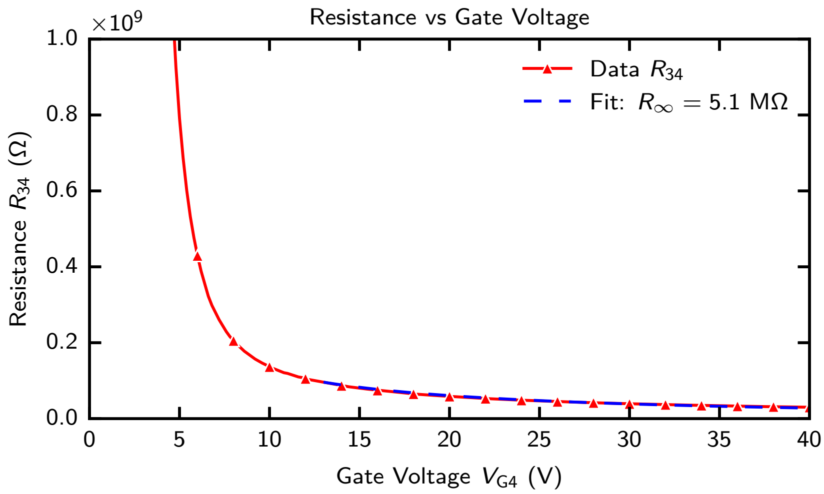

Figure 4.4: Drain–source resistance $R_{34} = V_{34}/I_{34}$ extracted from the low-drain-bias linear regime as a function of gate voltage. The resistance decreases as the negative gate bias accumulates a more conductive hole channel. The fitted background resistance represents the gate-independent part of the two-terminal resistance and is attributed mainly to injection/contact resistance associated with the Schottky barrier at the metal–organic semiconductor interface.

4.1.3 Output Characteristic

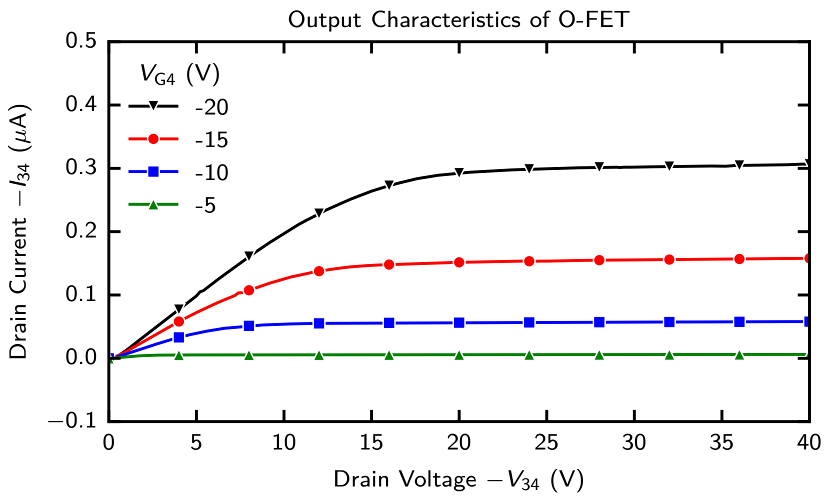

Figure 4.5: Output characteristic of the organic FET, measured as drain current $-I_{34}$ as a function of drain voltage $-V_{34}$ for different fixed gate voltages V_{G4}.

Figure 4.5 shows the drain current as a function of drain voltage for different fixed gate voltages. In this measurement, V_{G4} is fixed and the drain voltage V_{34} is swept. For $V_{G4} = -5\,\mathrm{V}$, the current remains close to zero, corresponding to a weakly accumulated or nearly off-state channel. For more negative gate voltages, the accumulated hole density increases and the output current becomes larger. The highest current is obtained at $V_{G4} = -20\,\mathrm{V}$.

At small drain voltages, the current increases approximately linearly. This indicates that the device forms a continuous conducting path and that the contacts inject holes into the organic semiconductor. At larger drain voltages, the curves bend over and approach a saturation-like regime. In a field-effect transistor, this occurs because the local gate-voltage overdrive decreases along the channel towards the drain electrode [12, 26]. Once the channel near the drain becomes strongly depleted, the current no longer increases linearly with drain voltage.

The output curves are smooth and ordered with gate voltage, which is a positive sign for the electrical stability of the device. Nevertheless, deviations from ideal transistor behaviour may still be present. In organic thin-film transistors, the shape of the output curves can be influenced by trap filling, field-dependent mobility, grain boundaries, and voltage drops at the metal–semiconductor contacts [2, 23, 3]. These effects are especially important when mobility is extracted from standard OFET equations, because the extracted value may describe the complete two-terminal device rather than the intrinsic semiconductor film.

Overall, the transfer and output characteristics show that the fabricated device behaves as a p-type OFET and that the gate voltage effectively modulates the current between contacts 3 and 4. At the same time, the limited current level and the possible influence of contact resistance show why a four-terminal method is needed. The gated van der Pauw method used in the following sections reduces the influence of injecting contacts and allows a more reliable extraction of sheet conductance, mobility, and threshold voltage [5, 7].

4.2 Gated Van-der-Pauw Method

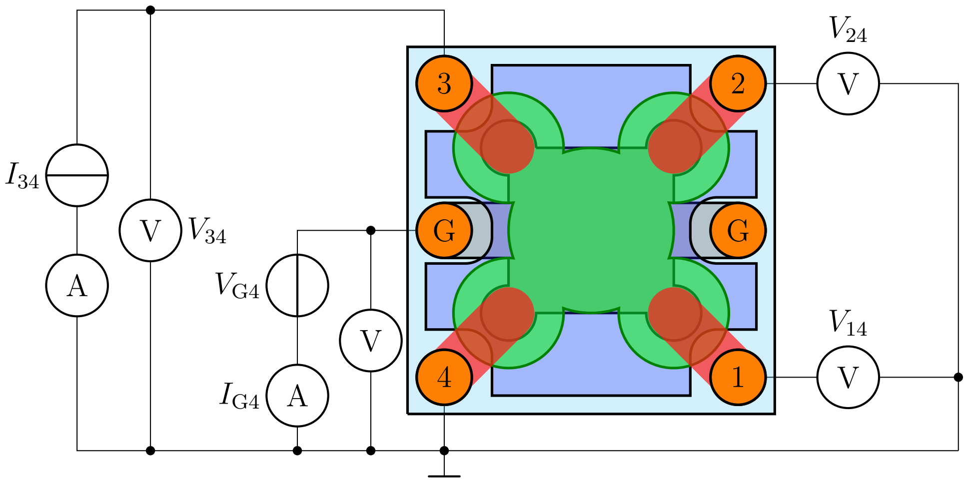

Figure 4.6: Gated van der Pauw measurement circuit used to extract the intrinsic sheet conductance. A constant current I_{34} is driven between contacts 3 and 4 while the voltage V_{34} is monitored. The transverse probe voltages V_{14} and V_{24} are measured with high-impedance voltmeters, and the gate branch applies V_{G4} while monitoring the leakage current I_{G4}. Adapted from Ref. [28].

4.2.1 Extraction of Intrinsic Mobility and Threshold Voltage

The gated van der Pauw measurement was used to separate the intrinsic sheet transport of the accumulated semiconductor film from the contact resistance of the injecting electrodes [5, 7]. A constant current I_{34} is driven through the device, while the voltage probes measure the transverse van der Pauw voltage with negligible current draw. From the measured sheet conductance σ, the intrinsic mobility and threshold voltage are obtained from the linear gate-voltage dependence of the conductance,

$$\sigma = \mu C_{\mathrm{i}} \left(V_{\mathrm{eff}} - V_{\mathrm{T,eff}} \right), \qquad \mu = \frac{1}{C_{\mathrm{i}}} \frac{\mathrm{d}\sigma}{\mathrm{d}V_{\mathrm{eff}}}, \qquad V_{\mathrm{eff}} = -(V_{\mathrm{G4}} - V_{\mathrm{C}}), \tag{4.1}$$

where $C_{\mathrm{i}} = 2\,\mathrm{nF}/32\,\mathrm{mm}^2 = 6.25\,\mathrm{nF/cm}^2$ is the gate-insulator capacitance per unit area. The slope of the conductance fit gives the intrinsic mobility, while the extrapolated intercept gives the threshold voltage.

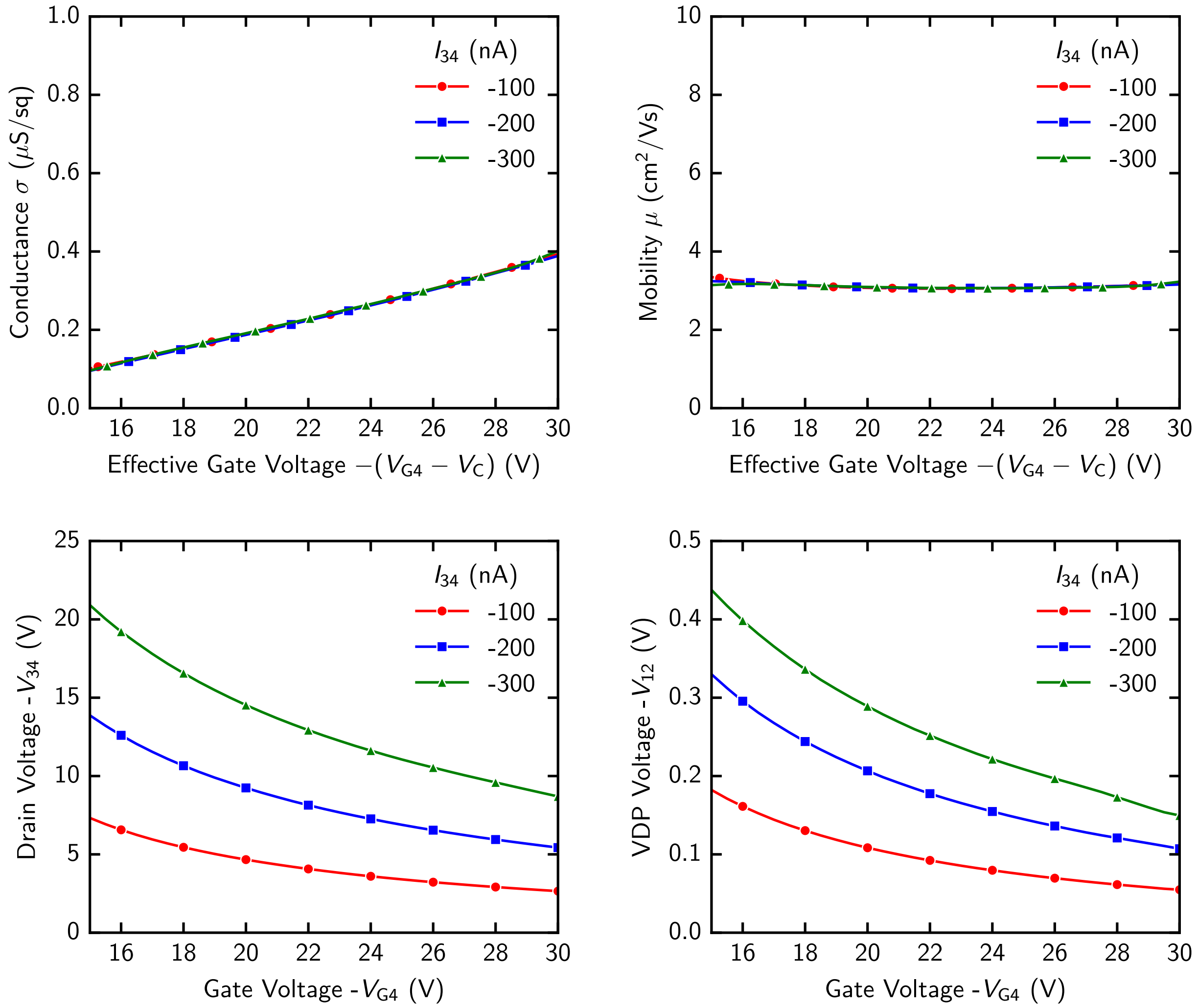

Figure 4.7: Gated van der Pauw extraction of the intrinsic sheet conductance, mobility, and threshold voltage for three applied currents, $I_{34} = -100\,\mathrm{nA}$, $-200\,\mathrm{nA}$, and $-300\,\mathrm{nA}$. The bottom panels show the measured drain voltage $-V_{34}$ and van der Pauw voltage $-V_{12}$ as functions of gate voltage. Both voltages decrease as the negative gate bias increases, because the accumulated channel becomes more conductive. The top-left panel shows the resulting sheet conductance σ as a function of effective gate voltage $-(V_{\mathrm{G4}} - V_{\mathrm{C}})$, and the top-right panel shows the mobility extracted from the conductance slope. The collapse of the data for different currents indicates linear-response behaviour and supports a current-independent extraction of both mobility and threshold voltage.

Figure 4.7 summarizes the complete extraction procedure. In the raw voltage data, the magnitude of $-V_{34}$ increases with the applied current, as expected from Ohm's law, but it decreases with increasing $-V_{G4}$ because the gate bias creates a denser hole-accumulation layer. The van der Pauw voltage $-V_{12}$ follows the same trend and is smaller than the longitudinal voltage because it probes the lateral sheet resistance rather than the full source–drain voltage drop.

After conversion to sheet conductance, the three current series collapse onto a single nearly linear curve. This current independence is important because it shows that the sheet conductance is not an artefact of the chosen current bias or of self-heating [5]. The linear increase of σ with effective gate voltage indicates that the induced sheet charge is approximately proportional to the gate overdrive in this voltage range. The extracted mobility is nearly constant, showing that the gated van der Pauw method yields a stable intrinsic mobility in contrast to two-terminal measurements, where contact resistance and injection barriers can strongly distort the apparent mobility [4, 3].

I_{34} (nA)	μ (cm^2/Vs)	V_T (V)	R^2
-100	3.11 ± 0.02	-10.16 ± 0.20	0.9981
-200	3.11 ± 0.02	-10.31 ± 0.16	0.9987
-300	3.10 ± 0.02	-10.11 ± 0.13	0.9991

Table 4.1: Fit results from the linear conductance analysis in Figure 4.7. The mobility is obtained from the conductance slope, and the threshold voltage is obtained from the extrapolated intercept.

The fit results are listed in Table 4.1. The mobility values agree within their uncertainties for all three applied currents, giving an intrinsic mobility of approximately $3.1\,\mathrm{cm^2/(V\,s)}$. The threshold voltages are also consistent, with values clustered around $-10.2\,\mathrm{V}$. The high coefficients of determination, $R^2 > 0.998$, confirm that the conductance is well described by a linear dependence on effective gate voltage over the fitted range. The small variation of both μ and V_T with current demonstrates that the extraction is robust and that the gated van der Pauw geometry successfully suppresses contact-related artefacts in the mobility and threshold-voltage analysis.

4.3 Additional Electrical Measurements

In addition to the current–voltage measurements discussed above, two further electrical tests were performed to probe time-dependent and circuit-level behaviour of the fabricated OFETs. The first test uses a deep-level transient spectroscopy (DLTS)-type experiment to study slow charge relaxation after a gate-bias perturbation [30, 9]. The second test evaluates whether the device can be biased and operated as a simple voltage amplifier. These measurements are complementary to the transfer and output characteristics: they do not only test the static channel conductance, but also reveal how traps, capacitances, contact effects, and external circuit elements influence the dynamic response of the organic transistor.

4.3.1 Deep-level Transient Spectroscopy

Organic semiconductors usually contain a broad distribution of localized states originating from energetic disorder, impurities, grain boundaries, and interfaces. These states can capture and release charge carriers on time scales that are much longer than the intrinsic carrier transit time [9, 10]. As a result, they can contribute to hysteresis, threshold-voltage shifts, slow current relaxation, and bias-stress effects. A DLTS-type transient measurement was therefore used to perturb the device electrically and observe the relaxation of the current after the perturbation.

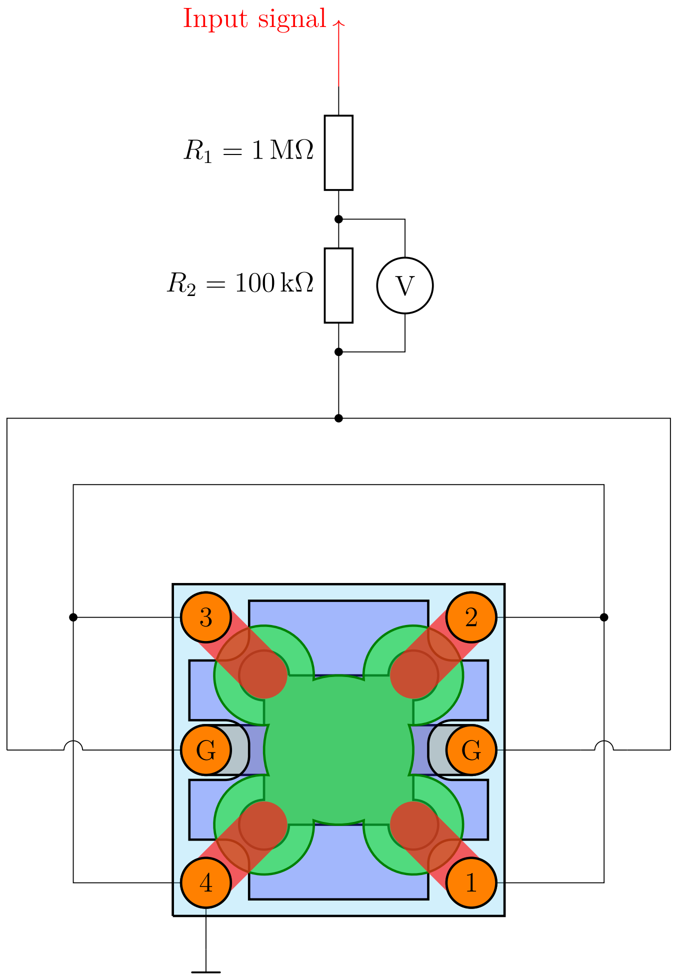

Figure 4.8: Circuit used for the deep-level transient spectroscopy (DLTS)-type measurements on the OFET structure. The external excitation is applied through the resistive divider formed by $R_1 = 1\,\mathrm{M\Omega}$ and $R_2 = 100\,\mathrm{k\Omega}$. The voltage across R_2 is monitored so that the transient response of the device can be followed after each bias perturbation.

The measurement circuit is shown in Figure 4.8. The input waveform is attenuated by the resistive divider before it is applied to the device. This configuration limits the current through the structure and provides a defined electrical perturbation of the gate bias. After a voltage step, the measured current contains an instantaneous capacitive contribution followed by a slower relaxation. The effective capacitance is obtained from the charge displaced during the transient,

$$Q = \int_{t_0}^{t_1} I(t)\,\mathrm{d}t, \qquad C_{\mathrm{eff}} = \frac{Q}{V_0} = \frac{1}{V_0}\int_{t_0}^{t_1} I(t)\,\mathrm{d}t, \tag{4.2}$$

where V_0 is the applied voltage after the divider and the current is integrated over the transient window after subtracting the long-time background. The remaining slow part is the relevant component for trap analysis, because it reflects the time required for localized states to exchange carriers with the conducting channel.

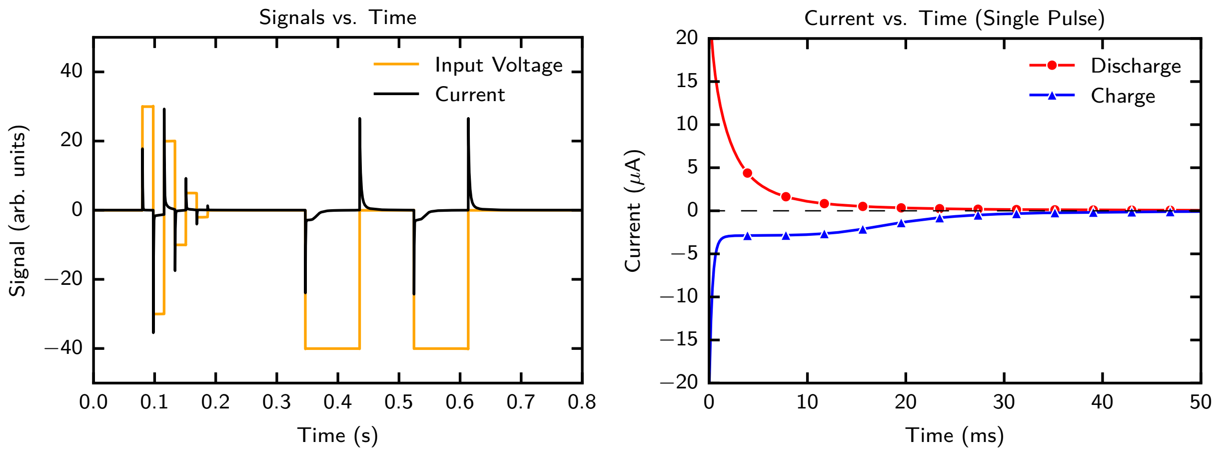

Figure 4.9: DLTS excitation waveform and representative single-pulse transient response. The left panel shows the applied voltage pulses together with the corresponding current response. The right panel compares the charge and discharge transients after one pulse, showing a fast initial response followed by a slower relaxation towards the baseline.

Figure 4.9 shows the pulse sequence and a representative single transient. The sharp current peaks at the voltage edges originate mainly from the rapid charging or discharging of the device capacitance. After this initial response, the current decays more gradually. The charge and discharge branches have opposite signs, which is expected because the electric field drives carriers into or out of trap states depending on the direction of the voltage step. The non-instantaneous return to the baseline indicates that the OFET response is not purely capacitive, but also contains slower trapping and detrapping processes.

For a qualitative interpretation, the transient current can be considered as a sum of relaxation components,

$$I(t) = I_\infty + \sum_i A_i \exp\left(-\frac{t}{\tau_i}\right), \tag{4.3}$$

where I_∞ is the long-time background current, A_i is the amplitude of a relaxation component, and τ_i is the corresponding characteristic time constant. A single exponential would indicate one dominant trap time scale, whereas a broader or multi-exponential decay indicates a distribution of trap depths and local environments [30, 31, 32]. The measured traces show a rapid initial drop followed by a longer tail, which is consistent with a broad trap distribution typical of disordered organic semiconductor films.

The full transient trace in Figure 4.10 shows that the current relaxation depends on the applied bias. Larger voltage steps produce larger initial currents because more charge is displaced at the semiconductor–dielectric interface. Integrating each transient gives the displaced charge Q, and division by the corresponding voltage step gives the effective capacitance. At longer times, the traces converge towards a small residual current, indicating that the device approaches a stable state after the trap population has partially re-equilibrated. The extracted capacitance values are all close to 1.7 nF, so the geometric and interfacial capacitance contribution is nearly constant over this voltage range. The remaining bias-dependent tail is therefore attributed mainly to the slow filling and emptying of localized trap states rather than to a change in the device capacitance itself.

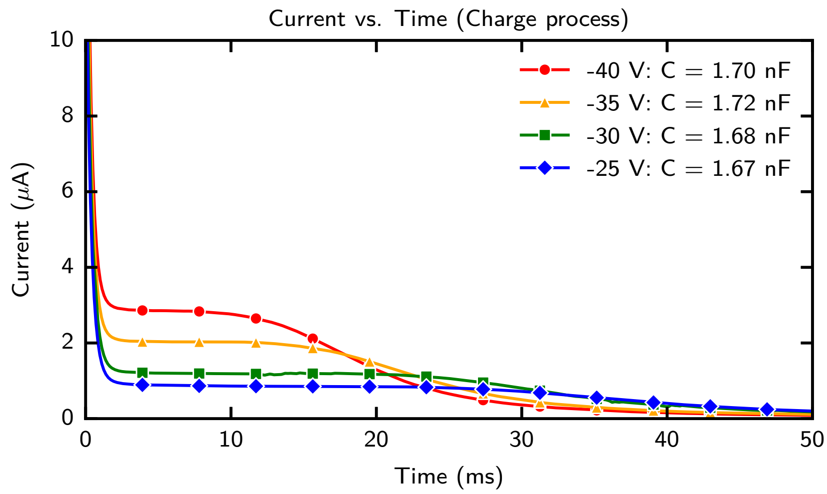

Figure 4.10: Full DLTS transient trace recorded during the charge process for different applied bias levels. The current relaxes after the voltage step and approaches a low background value. The effective capacitance is determined from $C_{\text{eff}} = Q/V_0$, with Q obtained by integrating the transient current. The extracted values remain close to 1.7 nF, indicating that the dominant fast contribution is capacitive, while the slower tail is associated with trap-related relaxation.

Overall, the DLTS-type experiment confirms that the OFET contains slow relaxation processes that are not visible from the static transfer curve alone. These processes can influence threshold-voltage extraction and apparent mobility when measurements are performed with different sweep rates or after different bias histories [10, 9]. This observation supports the use of careful bias protocols and contact-independent analysis methods in the following characterization.

4.3.2 Amplification Properties Analysis

The OFET was also tested in a simple amplifier configuration to evaluate whether the measured field-effect modulation can be used in an electronic circuit. The aim of this measurement is not to optimize a practical amplifier, but to demonstrate that changes in gate voltage can be converted into measurable changes in output voltage. This provides an additional, circuit-level confirmation of transistor action.

Figure 4.11 shows the common-source amplifier circuit used for this test. The gate bias defines the operating point of the OFET, and the drain resistor R_D converts a modulation of channel current into a voltage signal at the output. Because the device is a p-type transistor operated with a negative supply voltage, the sign convention differs from that of a standard n-channel amplifier, but the small-signal principle is the same: a change in gate voltage changes the channel current, and this current change produces a voltage drop across the load resistor [12, 33].

The static amplifier characteristics are shown in Figure 4.12. At small input voltages the channel is weakly accumulated, while at large input voltages it is already close to strong accumulation. The strongest gate control therefore occurs in the intermediate range, where the output voltage changes most rapidly. The gain is negative, as expected for a common-source stage, and increases in magnitude when R_D is raised from 20 MΩ to 50 MΩ. This

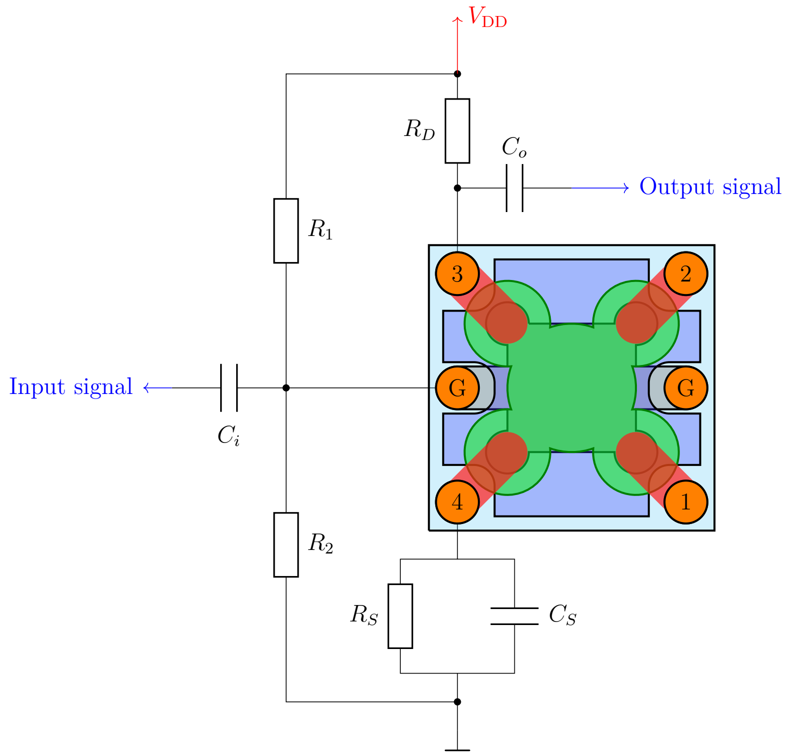

Figure 4.11: Circuit used to analyse the amplification properties of the OFET. The input signal is capacitively coupled to the gate through C_i, while the gate operating point is set by the bias divider R_1 and R_2. The drain is biased through R_D at $V_{DD} = -30\,\mathrm{V}$, and the output signal is measured at the drain contact through C_o. The source branch contains R_S and the bypass capacitor C_S, which set the source bias and influence the small-signal response.

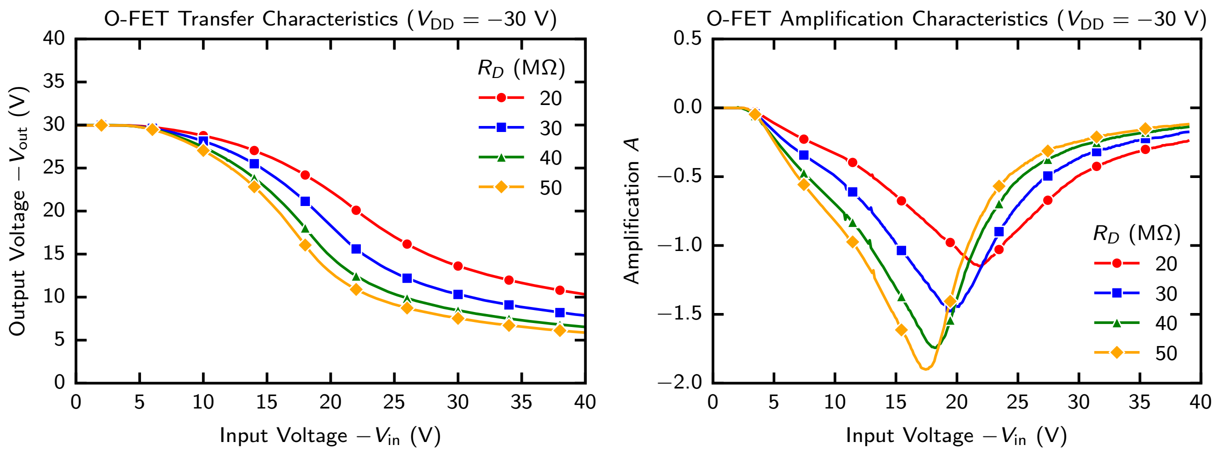

Figure 4.12: OFET transfer curve and amplification characteristic at $V_{DD} = -30\,\mathrm{V}$ for different drain-load resistances R_D. The transfer plot identifies the input-voltage range where the output voltage changes most strongly, while the amplification plot shows the corresponding small-signal voltage gain. Larger R_D values increase the maximum gain but also narrow the useful operating range.

larger load resistance also increases the influence of parasitic capacitances, so the highest-gain setting is not necessarily the fastest or most linear operating point.

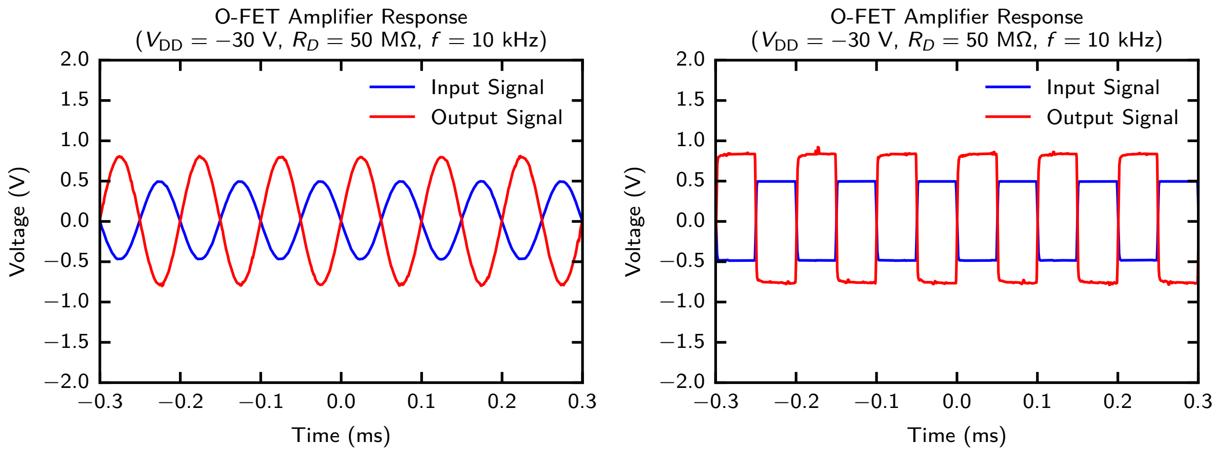

Figure 4.13: Dynamic response of the OFET amplifier at $V_{DD} = -30\,\mathrm{V}$, $R_D = 50\,\mathrm{M\Omega}$, and $f = 10\,\mathrm{kHz}$. The sinusoidal measurement demonstrates inverted voltage amplification, while the square-wave measurement shows the finite rise and fall times caused by the large circuit resistance, device capacitance, and trap-related relaxation.

Figure 4.13 shows the response to sinusoidal and square-wave inputs. The sinusoidal output is phase-inverted and larger than the input, confirming voltage amplification. The rounded waveform and the finite rise and fall times in the square-wave response show that the bandwidth is limited by the large load resistance, device and wiring capacitances, and trap-related relaxation [10, 9]. The output nevertheless follows the input reproducibly, so the OFET can operate as an active amplifying element under the chosen bias conditions.

Together with the DLTS results, this measurement shows that the devices combine useful field-effect modulation with significant dynamic effects. Contact effects, trap states, and time-dependent charging must therefore be considered when extracting quantitative parameters or designing circuits based on these OFETs.

CHAPTER 5

CONCLUSION AND SUMMARY

In conclusion, the aim of this thesis was to fabricate and characterize organic field-effect transistors in a geometry suitable for contact-independent transport analysis. The motivation was that conventional two-terminal OFET measurements do not probe only the semiconductor channel. Contact voltage drops, gate-dependent injection, trap-related relaxation, and possible leakage can all change the apparent slope of transfer curves and therefore distort extracted mobility and threshold voltage values [3, 4]. For this reason, the gated van der Pauw method was used as the main analysis tool, because it determines the sheet conductance from voltage probes that do not carry the imposed current [5, 7].

The theoretical discussion established the semiconductor and device-physics background needed for this comparison. It showed why ideal FET relations are only valid when contact effects, localized states, and leakage are sufficiently small, and introduced the gated van der Pauw method as a four-terminal alternative for measuring the accumulated sheet conductance. The fabricated devices used a bottom-gate architecture with a Cr gate electrode, a PaN/PS gate dielectric stack, a C8-BTBT molecular semiconductor layer, and MoO_x/Au injection contacts. The clover-leaf contact geometry made it possible to perform both conventional source–drain measurements and gated van der Pauw measurements on the same device type.

The two-terminal current–voltage measurements confirmed that the fabricated devices operate as p-type OFETs. Negative gate bias accumulated holes at the semiconductor–dielectric interface, the gate leakage remained small compared with the source–drain current, and the output curves showed the expected ordering with gate voltage. At the same time, the gradual turn-on of the transfer curves and the finite background resistance indicated that contact and injection effects remained important. Thus, the conventional measurements verified transistor action, but also showed that two-terminal mobility extraction alone is not sufficient for this device system.

The gated van der Pauw measurements provided the main quantitative result. Sheet conductance curves measured at $I_{34} = -100\,\mathrm{nA}$, $-200\,\mathrm{nA}$, and $-300\,\mathrm{nA}$ collapsed onto a common linear dependence on effective gate voltage. The extracted mobilities were $(3.11 \pm 0.02)\ \mathrm{cm^2/Vs}$, $(3.11 \pm 0.02)\ \mathrm{cm^2/Vs}$, and $(3.10 \pm 0.02)\ \mathrm{cm^2/Vs}$, with threshold voltages clustered around $-10.2\,\mathrm{V}$ and $R^2 > 0.998$ for all fits. The agreement between current levels shows that the extracted sheet conductance is not an artefact of the chosen drive current. The most reliable mobility obtained in this work is therefore approximately $3.1\mathrm{cm^2/Vs}$.

The additional electrical measurements complemented this static analysis. The DLTS-type experiment showed a fast capacitive response followed by slower current relaxation, consistent with localized states that exchange carriers with the channel over a distribution of time scales [30, 9, 10]. The effective capacitance remained close to 1.7 nF, while the relaxation tail indicated that traps and bias history can affect transfer characteristics. The amplifier test demonstrated inverted voltage amplification, but also showed that the dynamic response is limited by large circuit resistance, parasitic capacitance, and trap-related relaxation.

Overall, the fabricated C8-BTBT OFETs showed clear field-effect behaviour, but their two-terminal characteristics contained substantial contact and time-dependent contributions. The gated van der Pauw method separated the sheet-conductance measurement from the strongest contact voltage losses and yielded a stable, current-independent mobility. Within the scope of this thesis, the main objective was therefore achieved: a working OFET platform was fabricated, its nonideal two-terminal behaviour was identified, and a robust contact-independent mobility was extracted.

Several limitations remain. The number of devices and current levels was limited, so a broader statistical study is needed to separate device-to-device variation from systematic material behaviour. The numerical simulation used a simplified two-dimensional model and did not include microscopic injection physics, trap energetics, or the full gate-stack electrostatics. Future work should therefore combine gated van der Pauw measurements with temperature, contact-metal, sweep-rate, and device-statistics studies, and with more complete drift-diffusion–Poisson simulations that include contact injection and trap-state distributions.

APPENDIX A

CURRENT–VOLTAGE RELATIONSHIP OF FETS–MATHEMATICAL DERIVATION

This appendix gives the mathematical derivation of the ideal long-channel field-effect-transistor current–voltage relations used in Equation (2.37). The derivation follows the standard gradual-channel treatment of field-effect devices [12]. For consistency with the p-channel convention used in the main text, all voltages and currents are written as positive magnitudes.

The channel coordinate runs from the source at $x = 0$ to the drain at $x = L$. The local channel potential is $V(x)$, with

$$V(0) = 0, \qquad V(L) = V_{\mathrm{DS}}. \tag{A.1}$$

Within the gradual-channel approximation, the vertical gate field fixes the local accumulated sheet charge, while the much smaller lateral field drives the current. Above threshold, the mobile hole sheet charge is approximated by

$$Q_{\mathrm{p}}(x) = C_{\mathrm{i}}\left[V_{\mathrm{GS}} - V_{\mathrm{T}} - V(x)\right], \tag{A.2}$$

where C_{i} is the gate-insulator capacitance per unit area and V_{T} is the threshold voltage. The expression is valid only where the term in brackets is positive.

For constant mobility μ, the drift current through a channel of width W is

$$I_{\mathrm{D}} = W\mu Q_{\mathrm{p}}(x)\frac{\mathrm{d}V}{\mathrm{d}x}. \tag{A.3}$$

Since the steady-state current is independent of x, substitution of Equation (A.2) gives

$$I_{\mathrm{D}}\,\mathrm{d}x = W\mu C_{\mathrm{i}}\left[V_{\mathrm{GS}} - V_{\mathrm{T}} - V\right]\mathrm{d}V. \tag{A.4}$$

Integrating from source to drain yields

$$I_{\mathrm{D}}L = W\mu C_{\mathrm{i}}\int_0^{V_{\mathrm{DS}}}\left[V_{\mathrm{GS}} - V_{\mathrm{T}} - V\right]\mathrm{d}V \tag{A.5}$$

$$= W\mu C_{\mathrm{i}}\left[\left(V_{\mathrm{GS}} - V_{\mathrm{T}}\right)V_{\mathrm{DS}} - \frac{V_{\mathrm{DS}}^2}{2}\right]. \tag{A.6}$$

Therefore the ideal linear-regime current is

$$I_{\mathrm{D}} = \mu C_{\mathrm{i}}\frac{W}{L}\left[\left(V_{\mathrm{GS}} - V_{\mathrm{T}}\right)V_{\mathrm{DS}} - \frac{V_{\mathrm{DS}}^2}{2}\right], \qquad 0 \le V_{\mathrm{DS}} < V_{\mathrm{GS}} - V_{\mathrm{T}}. \tag{A.7}$$

For sufficiently small drain bias, the quadratic term can be neglected, giving

$$I_{\mathrm{D}} \approx \mu C_{\mathrm{i}} \frac{W}{L} \left(V_{\mathrm{GS}} - V_{\mathrm{T}}\right) V_{\mathrm{DS}}. \tag{A.8}$$

This limit shows that the accumulated channel behaves as a gate-controlled resistor whose conductance is proportional to $V_{\mathrm{GS}} - V_{\mathrm{T}}$.

The channel reaches the ideal pinch-off condition when the charge at the drain side vanishes,

$$Q_{\mathrm{p}}(L) = 0 \quad \Rightarrow \quad V_{\mathrm{DS,sat}} = V_{\mathrm{GS}} - V_{\mathrm{T}}. \tag{A.9}$$

Substitution of this condition into Equation (A.7) gives the ideal saturation current,

$$I_{\mathrm{D,sat}} = \frac{1}{2} \mu C_{\mathrm{i}} \frac{W}{L} \left(V_{\mathrm{GS}} - V_{\mathrm{T}}\right)^2. \tag{A.10}$$

In the ideal model this current is independent of further increases in V_{DS}. A finite output slope in real devices indicates additional nonidealities such as channel-length modulation, field-dependent mobility, contact resistance, or trap charging.

The usual mobility formulas follow directly from the two limiting forms. At small fixed V_{DS},

$$\mu_{\mathrm{lin}} = \frac{L}{W C_{\mathrm{i}} V_{\mathrm{DS}}} \frac{\mathrm{d}I_{\mathrm{D}}}{\mathrm{d}V_{\mathrm{GS}}}. \tag{A.11}$$

In saturation,

$$\sqrt{I_{\mathrm{D,sat}}} = \left(\frac{\mu C_{\mathrm{i}} W}{2L}\right)^{1/2} \left(V_{\mathrm{GS}} - V_{\mathrm{T}}\right), \tag{A.12}$$

and therefore

$$\mu_{\mathrm{sat}} = \frac{2L}{W C_{\mathrm{i}}} \left(\frac{\mathrm{d}\sqrt{I_{\mathrm{D}}}}{\mathrm{d}V_{\mathrm{GS}}}\right)^2. \tag{A.13}$$

These expressions are ideal model results. They give an intrinsic mobility only when the assumptions of a uniform long channel, constant mobility, negligible gate leakage, and negligible contact voltage loss are sufficiently fulfilled. Otherwise the extracted value should be interpreted as an apparent mobility.

APPENDIX B

DERIVATION OF THE GATED VAN-DER-PAUW RELATION

This appendix derives the relation between the measured sheet conductance and the gate-induced carrier density used in Section 2.4.3. The derivation follows the current-driven gated van der Pauw model introduced by Rolin et al. [5].

Consider the region between the two equipotential lines sensed by contacts 1 and 2. It can be represented by an equivalent field-effect channel with potentials V_1 and V_2. Since contact 4 is the reference in this thesis, $V_1 = V_{14}$ and $V_2 = V_{24}$. The local channel potential between these two lines is denoted by $V(x)$, where x is measured along the effective current path from the line at V_2 to the line at V_1:

$$V(0) = V_2, \qquad V(L) = V_1. \tag{B.1}$$

Within the gradual-channel approximation, the gate-induced sheet charge at this position is controlled by the voltage difference between the gate and the local semiconductor potential,

$$Q_{\mathrm{p}}(x) = C_{\mathrm{i}} \left[V_{\mathrm{G4}} - V_{\mathrm{T}} - V(x) \right]. \tag{B.2}$$

For constant mobility, the drift current through an effective channel width W is

$$I_{34} = W \mu Q_{\mathrm{p}}(x) \frac{\mathrm{d}V}{\mathrm{d}x}. \tag{B.3}$$

Substituting Equation (B.2) into Equation (B.3) and separating variables gives

$$I_{34}\, \mathrm{d}x = W \mu C_{\mathrm{i}} \left[V_{\mathrm{G4}} - V_{\mathrm{T}} - V \right] \mathrm{d}V. \tag{B.4}$$

Because the same current crosses every equipotential line, the left-hand side can be integrated from $x = 0$ to $x = L$, while the potential on the right-hand side is integrated from V_2 to V_1:

$$\begin{aligned} I_{34} L &= W \mu C_{\mathrm{i}} \int_{V_2}^{V_1} \left(V_{\mathrm{G4}} - V_{\mathrm{T}} - V \right) \mathrm{d}V && \text{(B.5)} \\ &= W \mu C_{\mathrm{i}} \left[\left(V_{\mathrm{G4}} - V_{\mathrm{T}} \right) \left(V_1 - V_2 \right) - \frac{V_1^2 - V_2^2}{2} \right]. && \text{(B.6)} \end{aligned}$$

Dividing by W and rewriting the bracket as a difference of squares gives

$$I_{34} \frac{L}{W} = \frac{\mu C_{\mathrm{i}}}{2} \left[\left(V_{\mathrm{G4}} - V_{\mathrm{T}} - V_2 \right)^2 - \left(V_{\mathrm{G4}} - V_{\mathrm{T}} - V_1 \right)^2 \right]. \tag{B.7}$$

The effective geometrical ratio of a symmetric van der Pauw structure is

$$\frac{L}{W} = \frac{\ln 2}{\pi}. \tag{B.8}$$

Using the difference of two squares in Equation (B.7) gives

$$\begin{aligned} I_{34}\frac{L}{W} &= \mu C_\mathrm{i}(V_1 - V_2)\left[V_\mathrm{G4} - V_\mathrm{T} - \frac{V_1 + V_2}{2}\right] && \text{(B.9)} \\ &= \mu C_\mathrm{i} V_{12}\left(V_\mathrm{G4} - V_\mathrm{T} - V_\mathrm{C}\right), && \text{(B.10)} \end{aligned}$$

where

$$V_{12} = V_1 - V_2, \qquad V_\mathrm{C} = \frac{V_1 + V_2}{2}. \tag{B.11}$$

The sheet conductance is defined by

$$I_{34}\frac{L}{W} = \sigma_\mathrm{s} V_{12}. \tag{B.12}$$

Equating Equations (B.10) and (B.12), and cancelling V_{12}, yields

$$\sigma_\mathrm{s} = \mu C_\mathrm{i}\left(V_\mathrm{G4} - V_\mathrm{C} - V_\mathrm{T}\right). \tag{B.13}$$

For a p-type accumulation layer, the magnitude is used,

$$\sigma_\mathrm{s} = \mu C_\mathrm{i}\left|V_\mathrm{G4} - V_\mathrm{C} - V_\mathrm{T}\right|. \tag{B.14}$$

Since $\sigma_\mathrm{s} = q n_\mathrm{2D}\mu$, comparison with Equation (B.13) gives the signed sheet carrier density

$$n_\mathrm{2D} = \frac{C_\mathrm{i}}{q}\left(V_\mathrm{G4} - V_\mathrm{C} - V_\mathrm{T}\right), \tag{B.15}$$

while the positive hole number density is the magnitude of this expression. Differentiation of Equation (B.14) with respect to the internal gate voltage gives

$$\mu = \frac{1}{C_\mathrm{i}}\left|\frac{\mathrm{d}\sigma_\mathrm{s}}{\mathrm{d}(V_\mathrm{G4} - V_\mathrm{C})}\right|, \tag{B.16}$$

provided that the mobility is approximately independent of gate voltage over the fitted interval. The derivation also shows directly why V_C must be subtracted: the accumulation charge is controlled by the voltage between the gate and the local semiconductor sheet, not by the gate voltage relative to the grounded current contact alone.

APPENDIX C

NUMERICAL SIMULATION OF THE VAN DER PAUW CONFIGURATION

This appendix summarizes the numerical procedure used to calculate the potential and hole-density maps shown in the theory chapter. The calculation is only a qualitative visualization of lateral electrostatics in a four-contact van der Pauw geometry. It is not used to extract a mobility, sheet density, or representative channel potential. Therefore the experimental correction V_{C} is not a relevant output of this numerical appendix; the voltage probe contacts are included only as passive geometrical markers.

The semiconductor is represented by a square two-dimensional grid with spacing h. The biased contacts impose fixed electrostatic potentials, while insulating outer boundaries satisfy zero-normal-field conditions. The local hole density is approximated by a Boltzmann relation,

$$p_{i,j} = N_{\mathrm{v}} \exp\left[-\frac{F_{i,j} + e\phi_{i,j}}{k_{\mathrm{B}}T}\right], \tag{C.1}$$

where $\phi_{i,j}$ is the electrostatic potential and $F_{i,j}$ is the local hole quasi-Fermi energy. The applied gate bias is not solved explicitly in this two-dimensional model; for a chosen V_{G4} it is represented by the fixed background charge density in Poisson's equation [24, 23],

$$\nabla^2\phi = -\frac{\rho}{\varepsilon}, \qquad \rho = e(p - N_{\mathrm{A}}), \tag{C.2}$$

Here N_{A} denotes this effective background charge density. In a full device model it may depend on the selected gate voltage, but in the present simplified Poisson solve it is held constant during the iteration.

For an interior grid point, the finite-difference form of Equation (C.2) is

$$\frac{\phi_{i+1,j} + \phi_{i-1,j} + \phi_{i,j+1} + \phi_{i,j-1} - 4\phi_{i,j}}{h^2} = -\frac{\rho_{i,j}}{\varepsilon}. \tag{C.3}$$

Solving this expression for the central value gives the Jacobi update candidate [34],

$$\phi^*_{i,j} = \frac{1}{4}\left(\phi^{(k)}_{i+1,j} + \phi^{(k)}_{i-1,j} + \phi^{(k)}_{i,j+1} + \phi^{(k)}_{i,j-1} + \frac{h^2\rho^{(k)}_{i,j}}{\varepsilon}\right). \tag{C.4}$$

Because p depends exponentially on ϕ, direct replacement can be unstable. The update is therefore under-relaxed, a standard stabilization strategy for nonlinear fixed-point iterations [34],

$$\phi^{(k+1)}_{i,j} = (1 - \alpha_\phi)\phi^{(k)}_{i,j} + \alpha_\phi \phi^*_{i,j}, \tag{C.5}$$

with $0 < \alpha_\phi < 1$. After each update, the contact potentials and insulating boundary conditions are reapplied.

The iterative solution is performed as follows. First, an initial potential $\phi^{(0)}$ satisfying the contact voltages is chosen. The hole density and space charge are then calculated from Equations (C.1) and (C.2). Next, all interior potentials are updated with Equations (C.4) and (C.5). This sequence is repeated until the relative change

$$\eta^{(k)} = \frac{\max_{i,j} |\phi_{i,j}^{(k+1)} - \phi_{i,j}^{(k)}|}{\max\left(\max_{i,j} |\phi_{i,j}^{(k)}|, 10^{-30}\right)} \tag{C.6}$$

falls below the selected tolerance. If current streamlines are required, the quasi-Fermi field is updated in a second conservative step from $\nabla \cdot (p\nabla F) = 0$, but the central electrostatic part of the calculation is the iterative Poisson solve described above.

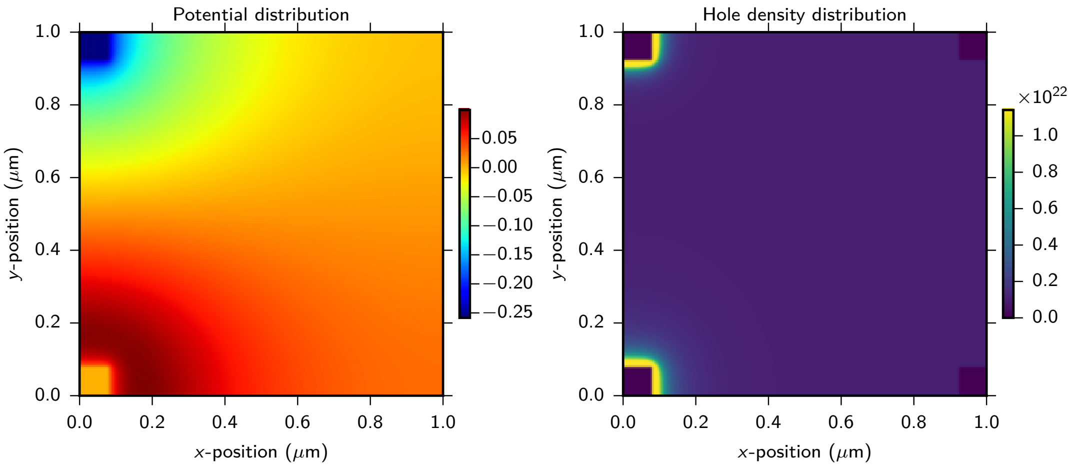

Figure C.1: Calculated potential and hole-density distributions in the gated van der Pauw structure. The potential map is obtained by iteratively solving Poisson's equation on the two-dimensional grid with fixed contact potentials and insulating outer boundaries. The hole-density map is then calculated from the local electrochemical energy and shows hole accumulation near the current-carrying contacts.

Figure C.1 illustrates the result of this iterative procedure. The potential is not uniform across the semiconductor sheet because the imposed contact potentials and insulating edges force the field lines to spread through the two-dimensional domain. Since the hole density is coupled to this potential through Equation (C.1), the density also becomes spatially nonuniform and is enhanced near the current-carrying contacts. The model remains idealized: it uses constant material parameters, Boltzmann statistics, simplified contact boundary conditions, and no explicit vertical gate-stack electrostatics. It should therefore be interpreted as a visualization of the iterative electrostatic solution rather than a quantitative prediction of the experimental mobility or carrier concentration.

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DECLARATION OF AUTHORSHIP

I, Trung Kien Pham, hereby declare that I have independently written the present thesis entitled "**Contact-Independent Electrical Characterization of Organic Thin-Film Field-Effect Transistors Using the Gated Van-der-Pauw Method**" and that I have used no sources or aids other than those indicated in the thesis.

Hiermit erkläre ich, Trung Kien Pham, dass ich die vorliegende Arbeit mit dem Titel "**Kontaktunabhängige elektrische Charakterisierung organischer Dünnschicht-Feldeffecktransistoren mittels der Gated Van-der-Pauw-Methode**" selbstständig verfasst und keine anderen als die in der Arbeit angegebenen Quellen und Hilfsmittel verwendet habe.

Munich, the 01.07.2026.
München, den 01.07.2026.

Trung Kien Pham